

Dynamic LLM Query Router (DLQR) for Multi-Model Orchestration



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Pranshu Goyal

School of Computer Science

University of Galway

Supervisor(s)

Matthias Nickles

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DECLARATION

I, Pranshu Goyal, hereby declare that this thesis, titled “Dynamic LLM Query Router (DLQR) for Multi-Model Orchestration”, and the work presented in it are entirely my own except where explicitly stated otherwise in the text, and that this work has not been previously submitted, in part or whole, to any university or institution for any degree, diploma, or other qualification.

Signature:

Name: Pranshu Goyal

Date: August 21, 2025

Acknowledgement We acknowledge the use of OpenAI’s ChatGPT for limited editorial suggestions and LaTeX formatting guidance[1]. Several aspects of the writing, language refinement and code comment in this thesis were assisted by ChatGPT (GPT-5 thinking) by OpenAI [1]. I have verified all generated content, checked formulas and results, and accept full responsibility for the final text and analyses. No automated system made independent research claims or final decisions.

Abstract

We evaluated the router and modules with transparent datasets, fixed splits, and a repeatable harness that measures discrimination, calibration, latency, throughput, and a token-aware cost proxy. The router achieves high routing accuracy with improved probability calibration and meets strict operational budgets (router p95 <100 ms, RAM <2 GB). End-to-end analysis attributes tail latency primarily to retrieval and generation, not routing. The measured results are reported in chapter 7; The results are computational evidence with auditable logs and risk controls aligned with the clinical AI reporting guidelines.

Keywords: orchestration, query routing, calibration, retrieval-augmented generation, case-based reasoning, selective prediction, latency, uncertainty

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Chapter 1

Introduction

1.1 Motivation and problem statement

Modern clinical decision support has to reason over mixed inputs: structured measurements (e.g. laboratory biomarkers) and unstructured narratives (e.g. clinicians’ questions and notes). Discriminative models perform well on tabular biomarker data, while large language models (LLMs) excel at free-text reasoning and retrieval-augmented answering. However, on their own, neither consistently optimizes discrimination, calibration, interpretability, latency, and cost across modalities [2, 3, 4, 5]. In cancer screening, minimally invasive body fluid biomarkers (saliva, oral rinse, blood) are maturing, motivating systems that combine predictive modeling with a textual clinical context [6, 7, 8].

We frame this as a systems-orchestration problem: for each input, route it to the most suitable expert—Prediction, Case-Based Reasoning (CBR), Clinical Q&A, or Summarisation—with calibrated confidence and the ability to abstain or defer under uncertainty. This draws on a mixture of experts and router designs from scalable AI [9, 10], and on recent LLM routing/cascades that balance accuracy against latency and cost [11, 12, 13]. Industry is converging on the same pattern:

OpenAI’s August 2025 release presents GPT-5 as “one unified system” with a real-time router that selects between non-reasoning and deeper-reasoning paths, trained on live preference and correctness signals [1, 14].

Threat model: Misrouting can degrade utility (e.g. sending structured biomarker queries to a generator) or increase cost/latency without benefit. DLQR mitigates these risks through post hoc calibration, explicit abstention thresholds, and conservative hedge policies that defer or reroute when uncertainty is high (cross-refer subsection 5.2.2, section 5.3; section 7.1 for reliability).

Our proposed Dynamic LLM Query Router (DLQR) inspects only text, the user query plus an optional short context string, to choose among these specialized modules. A compact policy head is calibrated with temperature scaling to produce reliable probabilities for thresholds, abstention, and fallbacks [15, 16]. For risk-aware operation, the router supports selective prediction (a rejection option) and can be paired with lightweight conformal methods, where appropriate [17, 18]. For auditability, we log decisions, confidence scores, latency, and evidential artifacts, aligning with contemporary reporting frameworks for AI routing (TRIPOD+AI, TRIPOD-LLM, DECIDE-AI) [19, 20, 21].

This is an AI/ML systems thesis that leverages medical-style data; it makes no clinical-effectiveness claims. All experiments are software-only. The DLQR is engineered to operate with p95 latency ≤ 100 ms and a ≤ 2 GB RAM footprint for the router component; end-to-end latency additionally reflects retrieval and generation in downstream modules.

Design drivers & positioning: Unlike prior cascades / routes that optimize only for accuracy or cost, DLQR is engineered under strict system constraints (router p95 ≤ 100 ms, ≤ 2 GB RAM) and limited router input (text only), and is evaluated against baselines of a single module (Always-Prediction, Always-QA) under matched hardware and concurrency (section 7.5).

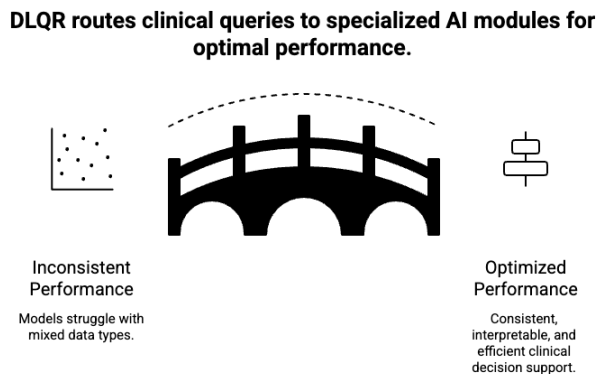


Figure 1.1: DLQR orchestrates AI experts.

1.2 Research questions

- RQ1: Does DLQR improve end-to-end diagnostic accuracy and efficiency (macro-F1/AUROC; p95 latency) versus single-model or naïve “one-size-fits-all” execution?
- RQ2: Does a dynamic LLM-assisted router improve latency and throughput compared to static ML or rule-based Routing?
- RQ3: How does adaptive learning (e.g., CBR case retention; contextual bandit-style updates to routing thresholds) affect routing performance over time?
- RQ4: What are the trade-offs among computer cost, accuracy and latency across Routing policies and fallback strategies?

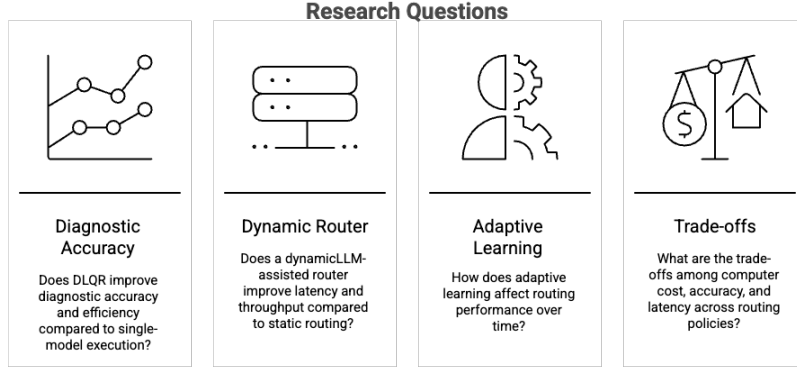


Figure 1.2: Research questions.

1.3 Contributions

DLQR calibrated with text only: We built a compact, text-only routing policy calibrated with temperature scaling ($\tau \approx 0.992$; mixture temperature $\tau_{\text{mix}} \approx 1.0$). It supports clear uncertainty thresholds and can abstain or defer when confidence is low. The policy can optionally incorporate a lightweight LLM intent prior to improving stability [15, 16].

Four-Module Orchestration: The system coordinates four interoperable modules: Prediction, Case-Based Reasoning (CBR), Clinical Q&A, and Summarisation behind consistent interfaces with shared explanation capabilities. For prediction we provide SHAP attributions; for CBR we show prototypes and nearest-neighbor rationales; and for Q&A we return retrieved evidence passages with citations in an RAG style [4, 5].

Transparent artifacts: We release all artifacts: synthetic and hybrid biomarker tables, the CBR case base, a text index with 30,636 chunks, the router corpus with fixed splits (6,656/2,048/1,536), calibration files, and decision logs.

Reproducibility specifics. We fix seeds/configs and release artifact hash IDs

for tables, indices, and logs to enable integrity checks and exact reruns.

1.4 Scope, assumptions, and ethics

Assumptions: (i) Routing observes only text; (ii) the Prediction module trains on de-identified tables; (iii) Q&A/Summarisation use retrieval-augmented generation with provenance; (iv) calibration and uncertainty govern abstention/fallbacks. Data are synthetic or hybridized/anonymized. Ethical, governance, and deployment considerations (bias, data protection, misuse) are discussed in chapter 7, chapter 8 together with a risk register. The reporting is aligned with TRIPOD + AI / TRIPOD-LLM for prediction studies and DECIDE-AI for the evaluation of early live systems [19, 20, 21].

Operational risks. Router latency is characterized under warmed caches; extreme cold starts/high concurrency can widen the end-to-end tail despite a bounded gate. Interpretability methods (e.g., SHAP) are model-dependent; retrieved evidence may omit confounders. These caveats limit claims to software-only conclusions (section 7.7).

1.5 Structure of the thesis

The thesis is organized into eight chapters. chapter 2 establishes the technical background, introducing calibration and uncertainty, selective prediction and deferral, specialist routing and orchestration, retrieval-augmented generation, and case-based reasoning (CBR). It also outlines the principles of biomarker-based prediction and reporting standards that guide the documentation of this work. chapter 3 reviews related research, synthesizing developments in routing paradigms such as cascading and cascade routing, bandit-style decision policies, and orchestration strategies in AI systems. This discussion develops a taxonomy

that clarifies existing approaches, identifies points of divergence, and highlights the gaps that motivate the DLQR design. chapter 4 describes the data used in the project, detailing the sources, schemas and construction of synthetic and hybrid tables parameterized by the literature. It also explains preprocessing, data splits, and leakage checks, while addressing variable dictionaries, class imbalance handling, and governance considerations to ensure that the pipeline is auditable and reproducible. chapter 5 sets out the methodology, presenting the design of the dynamic LLM Query Router (DLQR), the interfaces between modules, and the representations adopted for tabular and textual inputs. It further motivates the calibration and uncertainty strategy, specifies the inference-time decision policy (including abstention and fallback mechanisms), and justifies the implementation choices that enable the router to meet the stated efficiency constraints. chapter 6 outlines the experimental setup, articulating hypotheses and linking them to research questions, specifying evaluation metrics such as AUROC, macro-F1, p95 latency, throughput and token-based cost, and describing the protocols and baselines against which DLQR performance is assessed. chapter 7 presents measured results. Finally, chapter 8 provides discussion and conclusions, synthesizing answers to research questions, reflecting on limitations, ethics, and governance, and outlining directions for future work. The appendices provide supplementary material and links to the source code repository, supporting transparency and reproducibility of the study.

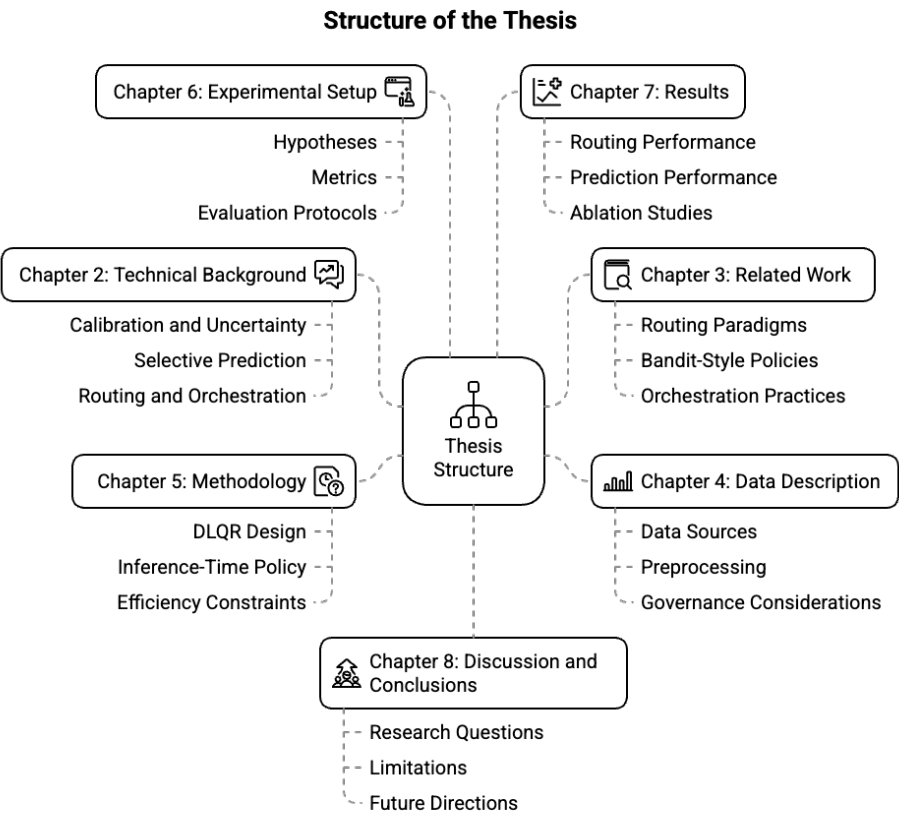


Figure 1.3: Thesis structure and components.

Chapter 2

Background

This chapter provides textbook-level context for the dynamic LLM Query Router (DLQR) and its expert modules. It establishes terminology, foundational methods, tooling, and evaluation practices so that the design choices in later chapters are clear. Where possible, we connect concepts with peer-reviewed literature, prioritizing the past ten years, and avoid clinical claims by focusing on computational aspects.

2.1 Conceptual map: modules, signals, and objectives

The DLQR orchestrates four specialized experts: a prediction model for structured biomarkers, a case-based reasoning (CBR) retriever for precedent-style analogies, a question answering (QA) module, and a summarization module. Each request is first represented as text (the user query plus a short optional context). The router encodes this text and produces a probability distribution over the targets. The highest confidence target is selected unless uncertainty controls suggest deferral or hedging. The end-to-end objectives balance accuracy / utility, p95 latency, and

compute/cost.

A clinician-style question such as "Considering these saliva biomarkers, what is the likely risk profile?" should go to prediction, while "What are the early symptoms to monitor?" routes to QA. A narrative case ("45 years old, tobacco exposure, persistent ulcer...") can be sent to CBR to retrieve similar cases and their results. The router's decision therefore aligns the task with the most appropriate expert while keeping latency and cost within the bound.

2.2 Intent classification and logit-space mixtures

Routing begins with intent classification: turning free text input into an operational intent such as biomarker risk, case similarity, or clinical question. Modern conversational systems rely on neural encoders for this step [22]. We use a Sentence Transformers encoder to obtain a compact embedding and a small multilayer perceptron (MLP) policy. For robustness, DLQR combines two signals: the learned policy logits and a lightweight intent prior derived from text cues. We mix them in logit space.

$$\text{logits}_{\text{mix}} = (1 - \alpha) \text{logits}_{\text{policy}} + \alpha \text{logits}_{\text{prior}}. \quad (2.1)$$

We then apply a softmax with mix temperature τ_{mix} :

$$p(a \mid x) = \text{softmax}\left(\frac{\text{logits}_{\text{mix}}}{\tau_{\text{mix}}}\right). \quad (2.2)$$

This preserves margins, exposes an explicit trade-off parameter α , and stabilizes routing when policy and the prior partially disagree.

2.3 Calibration and uncertainty: ECE, NLL, Brier

Neural networks tend to be over-confident. The temperature scaling multiplies the logarithmic by $1/\tau$ before the softmax, improving the probability calibration without changing the predictions of the top-1 [15]. Transformer classifiers also show calibration issues that vary by domain and prompt style [16]. Because calibration often deteriorates under distribution shift, we evaluated both independent and identically distributed (i.i.d. shifted conditions [23]. In DLQR, calibrated probabilities drive thresholding, abstention, and cost-aware routing decisions.

$$\tilde{\ell} = \frac{\ell}{\tau}, \quad p(y \mid x) = \text{softmax}(\tilde{\ell}), \quad (2.3)$$

where $\tau > 0$ is the temperature.

2.4 Selective prediction, deferral, hedging, and online adaptation

Selective prediction allows abstention in uncertain cases, trading protection for risk [24]. Learning to defer difficult instances to another expert (including a human), optimizing a combined objective on accuracy and deferral cost [25, 26]; see also a recent survey on reject options and post-training processing [17]. DLQR adopts these principles at the router level: the router either selects a target module or defers when calibrated confidence is below a threshold.

Ambiguous inputs benefit from hedging: issuing a selective top2 route so that two experts respond, then choosing the output with stronger internal evidence (e.g., CBR similarity vs. QA retrieval score). Over time, routing thresholds and mixture weights can adapt via lightweight contextual bandits; we use Linear Thompson Sampling (LinTS) over summary features (confidence, query length, task flags),

which balances exploration and exploitation under tight latency budgets [27]. Recent work on LLM routing/cascading shows that learned policies maintain quality while reducing cost [11, 12, 13].

2.5 Retrieval-augmented generation and evidence grounding

Retrieval-Augmented Generation (RAG) conditions generation on passages retrieved from a large corpus, anchoring outputs to explicit evidence, and mitigating hallucinations on knowledge-intensive tasks [4]. We combine BM25 (lexical) and dense passage retrieval, then apply a cross-encoder reranker to surface the most semantically relevant evidence [28, 29, 30]. The generator then conditions this context and is instructed to cite or echo sources. Evidence-based outcomes are essential in adjacent clinical settings.

2.6 Vector search and indexing at scale

Efficient retrieval requires a fast approximate nearest-neighbor search on large embedding stores. We use FAISS for CPU/GPU similarity search [31] and HNSW indices for low-latency, high-recall traversal on million-scale corpora [32]. Choosing between flat (maximum accuracy) and graph-based indices allows us to tune the recall-latency trade-off for task and hardware.

Index configuration (replication details): We use FAISS HNSW with $M=16$, $efConstruction=200$; query-time $efSearch \in \{32, 64, 128\}$ (default 64) balances recall and tail latency. For text embeddings, we store L2-normalized vectors; numeric vectors are z-scored per feature. We report latency at p50 / p90 / p95 per subpath (numeric / text) with the above settings (section 7.3, section 7.6)

2.7 Structured biomarker prediction and interpretability

The prediction expert consumes tabular biomarker features. Pre-processing includes imputation (median/mode or KNN), scaling (z score or min-max) and leak-robust splits. We evaluated XGBoost and a compact MLP baseline [33, 34]. Discrimination is assessed with precision, macroF1, AUROC, and AUPRC. Calibration is measured using ECE, NLL, and Brier. For interpretability, we compute SHAP values to attribute risk to features at global and local levels [35]. Biomarker-level evidence for oral cancer highlights inflammatory cytokines (e.g., IL6, IL8) and fragments such as CYFRA21 1, but heterogeneous cohorts and pre-analytical variation require careful validation [6, 7, 8]. In this paper, such features illustrate the model reasoning rather than support clinical claims.

2.8 Case-Based Reasoning (CBR) with hybrid similarity

CBR solves new problems by analogy with prior cases. We combine numeric similarity over scaled biomarker vectors with text similarity over embedded narratives (Sentence Transformers). The cases are grouped by k-means prototypes; at query time, the neighborhood of the nearest prototype is explored with kNN. A weighted fusion of numeric and text scores ranks neighbors, returning exemplars ("this resembles case X") that complement probabilistic predictions and provide precedent-style explanations. Recent healthcare care surveys document the complementary value of CBR in conjunction with model-based predictors [36].

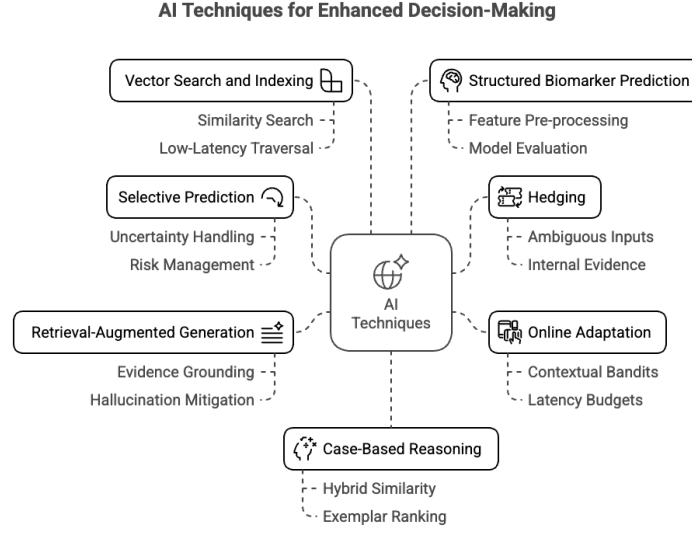


Figure 2.1: AI Techniques for Enhanced Decision Making

2.9 Generator model and prompt design

For QA and summarisation, we use a Mistral-class instruction-tuned LLM configured for efficient inference [37]. The prompts include retrieved passages, citation markers, and concise response instructions. When retrieval is weak, router calibration and thresholds favor rerouting to CBR or deferral rather than unsupported generation.

2.10 Efficiency instrumentation: latency, throughput, and cost

Because orchestration is a system problem, we measure both quality and performance. DLQR reports latencies p50 and p95 (typical and tail behavior), throughput (requests / s) and a token-weighted cost per query. Tail latency

control is essential for interactive systems [38]. Cost-aware routing follows current practice in multi-LLM systems [11]. The router is engineered to meet p95 100 ms and 2 GB RAM, reserving most of the budget for downstream experts.

2.11 Software stack (tools and core libraries)

The implementation uses a reproducible Python ecosystem: Python 3.11 [39]; Jupyter/IPython for literate notebooks [40]; NumPy arrays [41]; pandas dataframes [42]; Matplotlib plotting [43]; scikit-learn preprocessing/metrics/k-means [44]; PyTorch for MLPs [34]; XGBoost for gradient-boosted trees [33]; Sentence-Transformers embeddings [45]; Hugging Face Transformers for model loading/inference [46]; Pyserini/Anserini for BM25 [47, 48]; FAISS/HNSW for dense search [31, 32]; serialisation with joblib/pickle; configurations/logs via JSON/YAML [49, 50]; optional seaborn and SciPy for statistical plots/routines [51, 52].

For drafting and editing support, ChatGPT (GPT-5 Thinking) [1] was used to suggest prose rewrites and code comments; All outputs were human-verified for correctness and clarity.

2.12 Algorithms and training techniques

We adopt standard training components: Adam optimization, batch normalization, dropout, and early stopping of ReLU / GELU activations [53, 54, 55, 56, 57]. For calibration we use temperature scaling and also discuss isotonic regression, Dirichlet calibration, and Bayesian binning into quantiles (BBQ) as alternatives [15, 58, 59, 60]. For coverage guarantees, conformal prediction provides distribution-free uncertainty control [18]. Retrieval uses a hybrid search that combines BM25 and DPR, with re-ranking of the cross encoder to refine topk lists [29, 30]. The

approximate nearest neighbors employ HNSW when the recall-latency trade-offs warrant it [32].

For drafting and editing support, ChatGPT (GPT-5 Thinking) [1] was used to suggest prose rewrites and code comments; All outputs were human-verified for correctness and clarity.

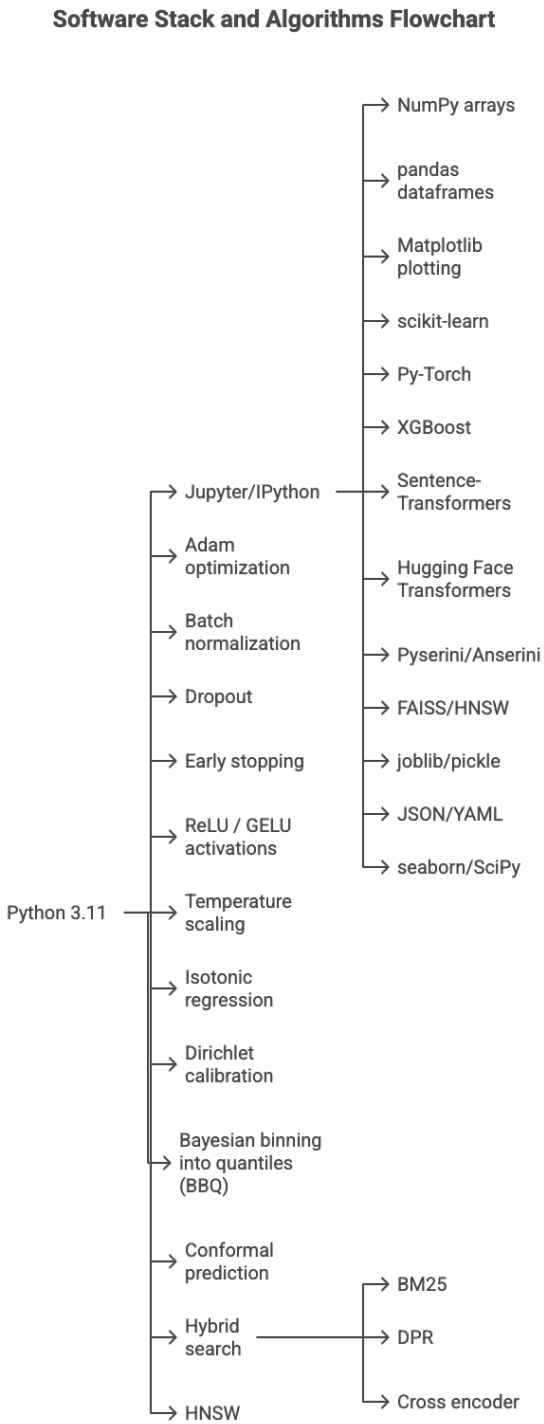


Figure 2.2: Software Stack and Algorithm Flowchart

2.13 Metrics, evaluation protocols, and statistics

We report accuracy, precision / recall / F1 (macro-averaged), AUROC for discrimination, and AUPRC for imbalanced regimes [61, 62]. Probability quality is measured using the ECE, the Brier score, and the NLL [15, 63]. Uncertainty is summarized with Wilson intervals for binomial proportions and bootstrap confidence intervals for AUROC/AUPRC/F1 and latency quantiles [64, 65]. Pairwise comparisons use the Wilcoxon signed rank test for nonparametric assumptions [66]. Latency distributions are analyzed via CDF and quantiles; throughput is reported as steady-state requests/s under fixed concurrency. A simple cost proxy combines latency and token usage to benchmark routing trade-offs [12].

2.14 Data assets and internal artifacts (reproducibility)

All data sets marked as synthetic or hybrid are generated by the authors. Structured tables include a large synthetic biomarker data set and a curated hybrid biomarker data set. CBR resources comprise a case library with train/test splits and an indexable store. Router resources include labeled router corpora (train, validation, and test), evaluation reports, and decision logs. Index files and model artefacts include FAISS indices, prototype clusters (`kmeans.joblib`), scalers/encoders, trained models (`xgboost_model.pkl`, `mlp_model.pth`), the router policy (`policy.pt`), and calibration files (`temperature.json`, `mix_temperature.json`).

A curated text corpus of 30,636 fragments of provenance evidence supports RAG. Where hybrid data or text are derived from public sources, original datasets and licenses are cited in chapter 4.

2.15 Governance, standards, privacy, and reporting

Although this thesis presents software-only results, we align with clinical AI reporting norms to frame limitations and avoid over-claiming. CONSORTAI and SPIRITAI cover trial reporting and protocol design for AI interventions [67, 68]. DECIDEAI addresses early-stage evaluations of decision support [19]. For prediction models, TRIPOD + AI updates the reporting for ML / AI [20]. For diagnostic accuracy, STARD 2015 and STARDAI inform the evaluation of transparency and bias [69, 70]. If personal data were to be processed, the EU GDPR principles would govern the lawful basis, minimization, and rights; the present work uses anonymized/synthetic data [71].

2.16 Industrial context: unified routing in production (2025)

Industry practice has converged on routing behind a single entry point. In August 2025, OpenAI described GPT5 as a unified endpoint supported by a real-time router that selects among the internal model families using signals about the type of task, complexity, and tool needs, balancing quality, latency, and cost [1, 14]. Our DLQR operationalizes this paradigm in a clinical adjacent setting with a transparent, compute-bounded router ($p95 \leq 100$ ms; ≤ 2 GB RAM) and reproducible software-only evaluations.

Chapter 3

Related Work

This chapter reviews the research strands that directly underpin the dynamic LLM Query Router (DLQR) implemented in this thesis. Rather than conducting an exhaustive survey, we synthesize methods that motivated our architectural choices and evaluation criteria.

3.1 Overview and taxonomy

DLQR operates at the intersection of three mature lines of work: (i) routing/cascading among heterogeneous experts to trade quality, latency, and cost; (ii) evidence-based generation via retrieval-augmented pipelines; and (iii) risk-aware decision making using calibration and selective deferral. Two other strands, structured biomarker prediction and case-based reasoning (CBR), provide non-generative experts whose behavior is transparent and computationally efficient. We organize related work accordingly and highlight where our system differs.

3.2 Routing and orchestration among experts

Sparse expert activation and conditional computation pioneered scalable mixtures with favorable cost / quality behavior [9, 10]. Building on this, LLM routing learns policies that choose between model families or depths: RouteLLM uses preference data to train routers that improve accuracy cost trade-offs [11], FrugalGPT explores cascades and back-off strategies to reduce spend while preserving quality [12], and a recent unification covers routing vs. cascading policies over heterogeneous models [13]. These trends are reflected in practice: OpenAI’s 2025 GPT5 announcements describe a single entry point backed by a real-time router that selects among internal model families based on task signals, balancing quality, latency, and cost [1, 14].

3.3 Evidence-grounded QA and summarisation (RAG)

For knowledge-intensive queries, Retrieval Augmented Generation (RAG) retrieves passages and conditions the generator on that evidence, improving factuality and reducing hallucinations [4]. Retrieval quality benefits from hybrid pipelines that combine BM25 lexical ranking with dense passage recovery (DPR) and encoder reclassification [28, 29, 30]. Scalable vector search is based on FAISS and graph-based HNSW indices to allow sublinear approximate nearest-neighbor search with low latency [31, 32]. DLQR adopts this pattern to constrain generative modules with provenance and to provide citations in outputs.

3.4 Structured biomarker prediction and interpretability

The tabular predictors remain strong baselines for risk stratification. XGBoost delivers competitive discrimination on structured clinical variables [33], and compact neural networks implemented in PyTorch serve as complementary baselines [34]. For interpretability, SHAP provides consistent attributions at global and local levels [35]. In oral cancer, recent reviews enumerate recurring salivary markers -IL6, IL8, CYFRA211, among others - while warning that cohort heterogeneity and pre-analytical variation complicate pooled estimates [6, 7, 8]. DLQR incorporates this strand by routing quantitative tasks to calibrated tabular models and using SHAP for explanatory plots; we refrain from clinical claims and focus on computational validity.

3.5 Case-Based Reasoning with numeric–text fusion

CBR addresses new problems by analogy to similar prior cases and is well established in healthcare [36]. Efficient retrieval combines prototype-based narrowing (e.g. k mean in numeric features) with an approximate nearest-neighbor search over dense text embeddings using FAISS/HNSW [31, 32]. In DLQR, CBR acts as a complementary expert for narrative input, returning precedent exemplars (e.g., nearest neighbors) as explanatory rationale alongside or instead of generative answers.

3.6 Calibration, uncertainty, and selective deferral

Reliable routing decisions require calibrated probabilities. Temperature scaling improves probability quality for deep networks—and for transformers specifically—without altering the predictions of the top 1 [15, 16]. In shifting data sets, predictive confidence decreases [72], so routers must be tested for stress beyond i.i.d. test data. When confidence is low, selective classification reduces risk by abstaining from uncertain cases [73]; learning to defer extends this by routing difficult instances to stronger experts (including humans) with explicit deferral costs [73, 23]. Reviews of rejection options summarize practical post-training tactics [17], and conformal prediction offers distribution-free coverage guarantees suitable for future extensions [18]. DLQR adopts these ideas to implement abstention and optional top2 hedging when two experts appear plausible.

3.7 Systems considerations: latency and cost

Interactive systems are dominated by tail latency; engineering for p95 is crucial [38]. Cost-aware orchestration reduces reliance on the most expensive models when simpler experts suffice [12]. Efficient retrieval through FAISS / HNSW [31, 32] and compact rerankers [39, 40] further reduces end-to-end latency. DLQR therefore measures p50/p95 latencies, throughput, and a token/latency cost proxy, and reports how routing choices affect each.

3.8 Evaluation practice and reporting standards

Methodologically, discrimination is reported with AUROC and AUPRC (for imbalance), while probability quality uses ECE, Brier score, and NLL; uncertainty intervals use Wilson and bootstrap methods; paired comparisons employ Wilcoxon

tests [15, 61, 62, 63, 64, 65, 66]. For transparency and governance in adjacent clinical research, we align with DECIDEAI for early stage evaluations, TRIPOD + AI and TRIPODLLM for LLM-based prediction and studies, and STARD 2015 / STARDAI for accurate diagnostic reporting [19, 20, 21, 67, 68, 69, 70]. The experiments use anonymized / synthetic data consistent with the GDPR principles [71].

3.9 Comparative positioning

System	Routing signal	Uncertainty handling	Evidence grounding	Domain focus	Compute budget
RouteLLM	Learned policy over LLM families	Implicit; quality via preferences	Optional (task-dependent)	General NLP	Varies; aims to reduce cost
FrugalGPT	Cascades/fallbacks by difficulty	Implicit via cascade thresholds	Optional	General NLP	Cost-minimising across models
Unified router (GPT-5)	Real-time router with task/tool signals	Production-grade heuristics (undisclosed)	Tool use + internal knowledge	Broad, production	Large-scale infrastructure
DLQR (this thesis)	Text-only calibrated policy + intent prior	Temperature-scaled thresholds; abstention/hedging; logs	Mandatory RAG for QA; SHAP/CBR for non-generative experts	Clinical-adjacent	Constrained (router p95 $\leq$ 100 ms; $\leq$ 2 GB RAM)

Table 3.1: Comparison of different routing systems across multiple dimensions.

3.9.1 Mechanism→Evidence mapping

Table 3.2: Techniques, their effects, citations, and usage within this work.

Technique	Effect / Rationale	Citations	Notes / Usage
Temperature scaling	Improves calibration (ECE/NLL/Brier) without altering accuracy meaningfully	[15, 59, 60, 63]	See section 7.1 reliability
Selective prediction / abstention	Reduces harmful decisions under uncertainty	[24, 25, 26]	Used for router deferral
Cost-aware routing/cascades	Maintains quality while cutting cost/latency	[11, 12, 13, 38]	DLQR aligns with budgets
FAISS/HNSW ANN	Low-latency, high-recall retrieval at scale	[31, 32]	Parameters in § 2.6
Evidence-grounded RAG	Reduces unsupported claims	[4, 5, 29, 30]	Gate in section 7.4

Critique temperature scaling under shift: While temperature scaling often reduces miscalibration in-distribution, distribution shift can re-introduce calibration error; this motivates considering optional conformal layers as distribution-free risk control.

3.10 Summary and gap

The literature shows that (i) routing / cascade improves the accuracy - latency - cost frontier [11, 12, 13]; (ii) RAG reliably grounds the generation of the retrieved evidence [4, 28, 29, 30]; and (iii) calibrated, selective policies reduce overconfident errors, especially under shift [15, 16, 23, 24, 25, 26]. The remaining gap is a computed, clinical adjacent orchestration layer that (a) performs text-only routing across heterogeneous experts (Prediction, CBR, QA, Summarisation); (b) enforces calibrated abstention/hedging with transparent logs; and (c) integrates evidence-based generation and interpretable non-generative experts under strict p95 latency budgets. DLQR targets this gap and provides a reproducible, software-only instantiation aligned with current reporting standards.

Chapter 4

Data

This chapter documents the datasets and internal artifacts used to develop and evaluate the dynamic LLM Query Router (DLQR). We describe sources, schemas, and labels; data quality, preprocessing, and leakage controls; splits and class imbalance handling; and finally privacy, governance, and known limitations. All experiments are software-only. Data are synthetic, hybrid, or deidentified and protected health information is not processed [71]. Where applicable, our reporting choices align with the DECIDE-AI, TRIPOD + AI / TRIPOD-LLM and STARD guidelines to avoid overclaiming beyond computational evaluation [19, 20, 21, 67, 68, 69, 70].

4.1 Data sources and acquisition

The study relies on four artifact families: (i) two tabular biomarker datasets for supervised prediction; (ii) a case-based reasoning (CBR) library that fuses numeric biomarkers with short clinical-style narratives; (iii) a text index of deidentified clinical-style passages for retrieval-augmented generation (RAG) and narrative similarity; and (iv) a router corpus of labeled queries for supervised routing. All

of these data sets were generated by the author, with no real patient records included [74, 75, 76, 77, 78].

The synthetic biomarker data set is the primary training table. It contains 100,000 rows and 74 columns covering cytokines (IL-6, IL-8), cytokeratin fragments (CYFRA 21-1), enzymes (ALP, LDH), metabolic markers, and selected microRNAs. divided in a triclass label $\text{Diagnosis} \in \{\text{Healthy Control}, \text{Benign Lesion}, \text{OSCC}\}$ and counts 39,985 / 30,126 / 29,889 respectively. The features cover inflammatory cytokines (e.g. IL-6, IL-8), cytokeratin fragments (e.g. CYFRA 21-1), enzymes (ALP, LDH), metabolic markers and a small set of engineered ratios. Synthetic generation allows controlled class balance and ablation experiments without privacy risk; we emphasize that these labels are computational training targets rather than clinical adjudications.

For robustness testing, a smaller hybrid biomarker table ($\approx 2,900$ rows) was parameterised directly from published group statistics. The mean, standard deviation, and sample sizes reported for biomarkers such as serum CYFRA 21-1, salivary IL-6, serum CRP, and salivary microRNAs (miR-31, miR-21) were extracted from peer-reviewed OSCC studies [79, 80, 81, 82, 83, 84, 85]. Using truncated Gaussian draws within physiological bounds, we simulated class-conditional distributions and retained missingness patterns similar to clinical data. All rows remain synthetic: no external patient-level records were ingested.

The CBR case library contains 10,000 cases. Each case pairs a 70 character biomarker vector with a short de-identified narrative (≤ 120 tokens) and a diagnosis label (OSCC = 4,091; Benign = 2,986; Healthy = 2,923). For efficiency, a 8,000-case subset forms the retrieval index, while 2,000 cases are left for the evaluation of nearest-neighbor reasoning.

The text index exposes 30,636 deidentified clinical-style passages, fragmented for retrieval, and embedded for dense vector search. It supports evidence grounding

for QA / summarisation through RAG [4], hybrid lexical–dense retrieval [28, 29, 30], and narrative similarity for the CBR text pathway [31, 32].

Finally, the router corpus comprises 10,240 short queries labeled with the intended module: prediction, cbr, qa, summarize. Labels are generated by templates and rules that reflect the remit of each module, so that a supervised router can learn the mapping from the query text to the target.

Table 4.1: Data sources and sizes

Artefact	Rows / Items	Columns / Fields	Labels	Notes
Synthetic biomarker table	100,000	74	Diagnosis $\in$ {Healthy, Benign, OSCC} = 39,985 / 30,126 / 29,889	Training; controlled missingness
Hybrid biomarker table	2,895	78	Diagnosis $\in$ {OSCC = 1,472; Healthy = 1,363; Benign = 60}	Evaluation; realistic missingness
CBR case library	10,000	79	As above	8,000 indexed; 2,000 hold-out
Text index (RAG/CBR)	30,636	—	—	De-identified clinical-style passages
Router corpus	10,240	—	RouteLabel $\in$ {prediction, cbr, qa, summarize}	Fixed splits

4.2 Data schema and dictionaries

The tabular artifacts share a consistent schema. The diagnosis label is categorical with three values (Healthy Control, Benign Lesion, OSCC). Numeric fields include IL-6 (pg/mL) and IL-8 (pg/mL) (inflammatory cytokines), CYFRA 21-1 (ng / mL) (a fragment of cytokeratin), enzymes such as ALP (U/L) and LDH (U/L), and systemic inflammation markers CRP (mg / L) and ESR (mm / hour). We also include Age (years) and Sex as optional covariates for tabular models; they are not used by the router. A small number of engineered ratios (for example, IL-6:IL-8 or ALP:LDH) provide simple normalisations that can improve discrimination without adding complexity. The choice of biomarkers is motivated by the recurring

appearances in the oral cancer literature; here they function purely as features for computational modeling rather than clinical assertions [6, 7, 8].

CBR cases mirror this numeric block and include a deidentified narrative. For the text index, each passage has a unique identifier, provenance tags, a fragmented text field (typically 256–384 tokens for stable retrieval context), and an embedding vector with metadata record encoder and dimensionality [45, 46]. The router corpus stores the query, an optional short context, a RouteLabel, and the split identifier.

Table 4.2: Variable dictionary (selected biomarkers)

Variable	Unit	Type	Rationale
IL-6	pg/mL	numeric	Inflammatory cytokine; appears in OSCC saliva literature [6, 7].
IL-8	pg/mL	numeric	As above [6, 7]
CYFRA 21-1	ng/mL	numeric	Cytokeratin fragment linked to head-and-neck malignancies [8]
ALP	U/L	numeric	Tissue turnover
LDH	U/L	numeric	Metabolic stress
CRP	mg/L	numeric	Systemic inflammation
ESR	mm/hr	numeric	Systemic inflammation
Age	years	numeric	Covariate (not used by router)
Sex	—	categorical	Covariate (not used by router)
IL-6:IL-8	—	numeric	Engineered ratio (post-split)
ALP:LDH	—	numeric	Engineered ratio (post-split)

Enumerates biomarker parameterization for the hybrid dataset (mean/SD, prevalence, cut-points) with exact literature citations per marker (e.g., IL-6, IL-8, CYFRA 21-1, ALP:LDH).

4.3 Label construction and ground truth

All labels in this study are simulation artifacts rather than clinical judgements. The rows in the biomarker tables were sampled from class-conditional distributions for Healthy, Benign, and OSCC, so the attached label simply reflects the process used to generate the case. In the synthetic tables and the CBR library, labels were programmatically assigned to preserve class balance and mimic plausible biological patterns, for example, elevating several inflammation markers together in cases labeled OSCC. The hybrid table instead inherited its labels directly

from the published sources on which its parameters were based, without any re-adjudication.

For unstructured text, deterministic templates and rules were used to decide whether a query sought risk prediction, case similarity, evidence-based answers, or a summary. In all settings, labels provide a consistent ground truth for training and evaluation, but must not be read as clinical diagnoses. They enable controlled tests of routing, calibration, and abstention strategies while the thesis itself does not make a claim of diagnostic performance, according to the clinical guidelines for reporting AI [19, 20, 21].

4.4 Data quality, missingness and leakage checks

Realistic missingness is a key property of the hybrid table; in contrast, the synthetic table injects controlled missingness so that imputation strategies can be stress-tested. Outliers are handled conservatively by winsorisation, which caps extreme values at a high/low percentile defined via the median absolute deviation (here $\text{median} \pm 5 \times \text{MAD}$), thereby keeping observations while reducing the leverage of aberrant readings. Strongly right-skewed features receive a $\log_{1p}$ transform to improve model stability. To prevent data leakage, all scalers, imputers, and encoders are fitted on the training split only; engineered ratios are created after splitting; nearest-neighbor structures and retrieval indices are built exclusively from training data per split; and no row appears in more than one split [44]. We also validate the integrity of the schema (types, ranges, units), class counts, and embedding shapes before training [41, 42].

Table 4.3: Missingness and imbalance (summary)

Dataset	Missingness (approx.)	Class balance notes
Synthetic biomarker	Injected (controlled)	Near-balanced
Hybrid biomarker	Moderate to high (feature-wise)	Benign $\approx 2\%$ (severe imbalance)
CBR library	Minimal (numeric); none (text)	Similar to biomarker distributions
Router corpus	N/A	Balanced test by design

4.5 Preprocessing and indexing

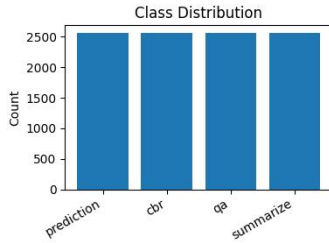
Numeric preprocessing follows standard practice with explicit guardrails. Continuous variables are imputed with the median; categoricals use the most frequent value. We also performed K-NN imputation ($k = 5$) as a sensitivity check to ensure that the results were not overly dependent on a single imputation scheme. All numeric features are scaled to z (mean 0, standard deviation 1) and $\log_{1p}$ is applied when the skew exceeds unity [44, 52]. Engineered ratios are computed post-split to avoid leakage.

Text processing performs lowercasing, light punctuation trimming and rule-based de-identification; passages are chunked to 256–384 tokens to balance retrieval specificity with context sufficiency. We compute sentence-level embeddings using Sentence-Transformers (all-MiniLM-L6-v2) and store them with model/version metadata [45, 46]. Downstream retrieval for QA uses a hybrid pipeline: BM25 for lexical matching, DPR for dense recall, and a cross-encoder reranker for final ordering [28, 29, 30]. The dense search is executed through FAISS; we deploy a flat index (higher recall) or HNSW (graph-based, lower latency) depending on the experiment [30, 31]. Numeric CBR also uses prototypes of k -means ($k = 6$, seed = 42) to narrow the search region before looking for the nearest neighbor [44].

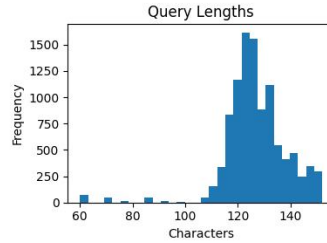
4.5.1 DLQR Preprocessing

Table 4.4: Preprocessing and indexing pipeline

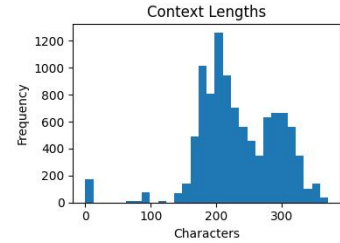
Step	Structured tables	Text index / CBR narratives	References
Missing value handling	Median (cont.); most-frequent (cat.); K-NN checks	—	[44, 52]
Outliers & skew	Winsorise ($\pm 5 \times \text{MAD}$); log1p if skew > 1	—	[44, 52]
Scaling	z-scaling (mean 0, sd 1)	L2 normalise embeddings	[44, 45]
Embedding	—	Sentence-Transformers (MiniLM)	[45, 46]
Retrieval	—	BM25 + DPR + cross-encoder rerank	[28, 29, 30]
ANN index	—	FAISS Flat / HNSW	[30, 31]
Prototypes	k-means (k = 6) for CBR numeric	—	[44]



(a) Class distribution after cleaning. Labels are near-balanced across the four routing intents, reducing bias risk and supporting stratified train/val/test splits.



(b) Query length histogram (characters). Most queries cluster around 120–130 characters, informing tokenizer limits and truncation settings for the intent head.



(c) Context length histogram (characters). Contexts tend to be longer (peak ~200–240) with a tail to ~350+ and a small mass at 0 (missing contexts).

Figure 4.1: Pre-processing audit plots used to set cleaning and length clamps. Balanced labels support robust stratified evaluation; observed length profiles guided max-length choices and safe truncation without altering semantics.

4.6 Splits, sampling and reproducibility

All supervised components use deterministic splits with fixed seeds. For prediction (synthetic), we apply a stratified 70/15/15 split (70,000 train; 15,000 validation; 15,000 test). The prediction (Hybrid) uses the same proportions (2,027 / 434 / 434). The CBR library is divided into 8,000 indexed cases and 2,000 holdout cases; the retrieval base never contains evaluation items. The router corpus uses fixed splits of 6,656 / 2,048 / 1,536 for train/validation/test, with balanced per-class counts in the test set. Pipelines are version-pinned (Python, NumPy, pandas, scikit-learn, PyTorch, Transformers), and artifacts are serialized with checksums for traceability [34, 39, 41, 42, 44, 46].

Table 4.5: Splits and leakage controls

Dataset	Train	Validation	Test	Leakage Controls
Prediction (Synthetic)	70,000	15,000	15,000	Fit transforms on train; engineer post-split; no row overlap
Prediction (Hybrid)	2,027	434	434	As above; conservative evaluation
CBR library	8,000 (index)	—	2,000 (hold-out)	Separate retrieval base vs evaluation
Router corpus	6,656	2,048	1,536	Fixed split IDs; balanced test per class

4.7 Class imbalance handling

The hybrid table is severely imbalanced for the Benign class (2%). Therefore, we train with class-weighted losses and balanced batch sampling, and we tune decision thresholds on the validation split (for example, using Youden’s J statistic) to improve minority-class recall without inflating false positives. Reporting emphasizes macro-F1 and class-wise metrics, which are more informative than accuracy in imbalanced regimes [61, 62]. Router training samples classes evenly to avoid degenerate ‘always choose X’ policies.

4.8 Privacy, governance and retention

In this study protected health information (PHI) is not used. All datasets were generated by the author [74, 75, 76, 77, 78], with published studies consulted only to provide statistical parameters such as means, standard deviations, and sample sizes. Because real patient records are not included, there is no risk of reidentification or discrimination.

Project artifacts, such as model files, scalars, and retrieval indices, are stored with versioning and cryptographic checksums under restricted access. This ensures integrity and aligns with the GDPR principle of data minimization, which requires collecting only what is strictly necessary [71]. As the experiments are software-only, participant consent was needed; however, results are reported using recognized frameworks (DECIDE-AI, TRIPOD, TRIPOD-LLM and STARD / STARD-AI) to maintain transparency and avoid overstating clinical relevance [19, 20, 21, 67, 68, 69, 70]. Any future work with real data would follow full GDPR requirements, including data subject rights.

4.9 Limitations and dataset shift

Our dataset construction reproduces broad statistics from oral squamous cell carcinoma (OSCC) studies, yet it cannot fully capture the finer dependencies seen in real patients. Approximating covariance in blocks makes computation tractable, but may exaggerate or miss correlations between biomarkers measured in different fluids or on different laboratory platforms. Likewise, published studies often suffer from small cohorts, selective reporting, or unbalanced stage distributions. These biases mean that the parameters we use may not generalize perfectly and the DLQR may respond differently when faced with data outside of these assumptions [23].

Synthetic data bring scale and control, but inevitably miss subtle preanalytical variation or assay-specific effects. The hybrid dataset, built by aggregating across multiple studies, can embed batch effects despite standardization. Labels are taken directly from their sources, so some noise is unavoidable. The clinical-style text index further adds realism, but is not site-specific, and retrieval-augmented generation is only as reliable as the material it can retrieve [4, 28, 29, 30].

To address these risks, we report not only discrimination metrics, but also calibration, abstention, and robustness measures (chapter 6, (chapter 7)). The results should therefore be read as evidence of computational feasibility, rather than as immediate proof of clinical utility.

4.10 Risks and Mitigations

Several risks were identified during the construction of the data set and the evaluation of the model. Parameter bias was an early concern, since the means and standard deviations often came from small or single-center studies. To reduce this, larger or more recent cohorts [80, 81, 82] were preferred, and heterogeneity was preserved by sampling in multiple studies. A related issue was scale mismatch, since biomarkers such as microRNAs and proteins are reported on different units (Ct/ Δ Ct, ng/mL, pg/mL). Rather than forcing them into a single scale, the pipeline retained native units with unit-aware transforms and clear documentation (section 4.2).

Another challenge was distributional shift: models trained and tested on matched simulations can appear over-optimistic. This was countered by focusing on calibration and abstention metrics and testing the robustness to change in later chapters [23]. To avoid data leakage, all transforms were fitted only to the training data, with ratios and retrieval indices computed after splitting, verified

both automatically and by spot checks [44].

Finally, the system design posed operational risks. Dense retrieval could inflate tail latency, so approximate nearest-neighbor indices and small top k values were used while tracking both median and p95 response times [31, 32, 38]. The class imbalance was handled with weighting, balanced sampling, and macro-averaged metrics [61, 62]. Generation modules were limited by retrieval-augmented generation and a calibrated router that could abstain when confidence was low [4, 15, 16].

These safeguards do not remove uncertainty entirely, but ensure performance claims rest on a transparent and robust foundation.

Chapter 5

Methodology

This chapter presents the Dynamic LLM Query Router (DLQR) as a systems methodology for selecting, at inference time, the most suitable expert among Prediction, Case-Based Reasoning (CBR), Question Answering (QA), and Summarisation, given a clinical-style text query. The design objectives are explicitly computational: router-only latency below 100 ms at the 95th percentile, a memory footprint under 2 GB, calibrated probabilities to enable thresholding, abstention, and deferral, and cost-aware orchestration that balances expected benefit with inference overhead. Generation is enhanced through retrieval-augmented methods, which mitigate hallucinations and ensure grounding in evidence [4, 5]. Methodologically, the work emphasizes calibration before action [15, 16, 59, 60], selective prediction and learning to defer under high uncertainty [24, 25, 26], and cost-latency-aware routing strategies [11, 12, 13], while avoiding tail-latency surprises [38]. The principles of transparency and scope control follow emerging guidelines such as DECIDE-AI, TRIPOD+AI, and TRIPOD-LLM [19, 20, 21].

5.1 Modules and their interfaces to the router

5.1.1 Prediction (structured biomarkers)

The prediction module is designed to assign each input case to one of three diagnostic categories: healthy, benign, or oral squamous cell carcinoma (OSCC). Its inputs are tabular biomarker vectors derived from the synthetic and literature-parameterised datasets described in chapter 4. Vectors represent a variety of biomarkers that have been reported recurrently in the oral cancer literature, including inflammatory cytokines such as interleukin-6 (IL-6) and interleukin-8 (IL-8), cytokeratin fragments such as CYFRA 21-1, enzyme indicators such as alkaline phosphatase (ALP) and lactate dehydrogenase (LDH), and systemic inflammatory markers such as C-reactive protein (CRP) [6, 7, 8, 33]. Although these biomarkers have clinical relevance in previous studies, in this work they are strictly treated as computational features that provide informative signals to the model, rather than as definitive diagnostic tools [6, 7, 8, 79, 80, 81, 82, 83, 84, 85].

The model architecture combines two complementary approaches. First, we employ XGBoost, a gradient-boosted decision tree algorithm that efficiently captures non-linear interactions between heterogeneous features and is known for its strong performance on tabular data. This is paired with a lightweight multilayer perceptron (MLP) implemented in PyTorch, which is better suited to model smooth, distributed patterns across continuous biomarker values [33, 34]. The two models together offer both robustness and flexibility, balancing the strengths of tree-based learning with those of neural architectures. To support interpretability, an essential requirement in a clinical-style context, we apply SHAP (Shapley Additive Explanations), which provides both global summaries of feature importance and local explanations for individual predictions. These explanations not only help identify potential model failure modes, but also help

align the output with established domain intuition [35].

To ensure that module output can be meaningfully integrated into the dynamic LLM Query Router (DLQR), we calibrate the raw softmax probabilities using temperature scaling. This post hoc adjustment corrects the tendency of models to produce over- or under-confident probabilities, improving the correspondence between predicted probabilities and empirical frequencies [15, 16, 59, 60]. The calibrated outputs serve as the primary signals to the router. Specifically, the module provides the DLQR with the top-class prediction along with its calibrated confidence, an optional margin representing the probability gap between the most likely and second most likely classes, and a compact digest of the most influential SHAP features. Although the digest is not used directly for routing, it is stored for logging and inspection purposes, ensuring that routing decisions can later be interpreted and validated. This design maintains both computational efficiency and transparency, allowing the prediction module to act as a reliable and explainable expert within the larger DLQR system.

AI tool disclosure. ChatGPT (GPT-5 Thinking) by OpenAI [1] was used only for the refinement of the prose and commenting, analysis, no statistical tests or results were delegated.

5.1.2 Case-Based Reasoning (CBR)

The Case-Based Reasoning module is intended to provide a different kind of support to the user than a categorical prediction. Instead of returning a label with probabilities, it looks for precedent: it retrieves earlier cases that resemble the current query and presents them alongside a short rationale. The aim is to encourage reasoning by analogy—'this case looks like those', which is often more intuitive and persuasive than a probability score alone, particularly in clinical and biomedical contexts where precedent plays a central role in decision making [36].

The numeric retrieval pathway operates on biomarker vectors that have been normalized to allow a fair comparison between features. These vectors are grouped into representative prototypes by clustering k-means. When a new query is introduced, the system identifies the nearest prototype and then retrieves its candidate neighbors from within that region. This structure makes retrieval computationally efficient and improves robustness, since the cases are drawn from the most relevant local subspace rather than from the dataset as a whole. By anchoring retrieval in this way, the system reduces the risk of spurious matches and ensures that retrieved analogies remain clinically plausible [44].

In parallel, the textual retrieval pathway handles short narrative descriptions, such as clinical notes or summary of case histories. These are embedded into dense vector representations using Sentence Transformers, which capture semantic meaning beyond the surface language [45, 46]. To keep query time fast even over large repositories, the embeddings are indexed and searched with FAISS, using flat indices or HNSW graphs depending on latency and memory trade-offs [31, 32]. This combination allows for an efficient and scalable approximate nearest-neighbor search, ensuring that narratives with semantically similar content can be retrieved in near real time.

When both structured and unstructured inputs are provided, the module fuses these two signals through a weighted blend of numeric and textual similarity scores. The top ranked cases are then returned, each accompanied by a short explanation that highlights overlapping biomarker features or shared phrases in the narratives. These rationales are central to the CBR philosophy, as they enable the user to understand not only which cases were retrieved, but also why they were deemed relevant [36].

Finally, the module communicates its output to the Dynamic LLM Query Router. Specifically, it provides an aggregate retrieval score, typically the average

similarity across the top k results, as well as a coverage flag indicating whether sufficiently close cases were identified. These signals allow the router to judge whether the query is well suited to CBR or whether an alternative module, such as prediction or summarisation, should be prioritized.

AI tool disclosure. ChatGPT (GPT-5 Thinking) by OpenAI [1] was used only for the refinement of the prose and commenting, analysis, no statistical tests or results were delegated.

5.1.3 QA (evidence-grounded) and Summarisation

The Question Answering (QA) and Summarisation module addresses information-seeking tasks where the user requires either a direct answer grounded in evidence or a condensed reformulation of provided material. The QA pathway is explicitly constrained to retrieval-augmented generation (RAG), which ensures that answers are drawn only from supporting documents rather than generated freely. This safeguard is important, as it reduces the risk of false claims and keeps the results tied to the underlying evidence base [4, 5]. In contrast, the summarisation pathway is designed as a safe fallback. Condenses or rewrites the input into structured prose when the system is uncertain about a categorical response or when the user explicitly requests a synthesis. In both roles, the module aims to balance information reliability with reliability, supporting knowledge-seeking queries without straying beyond the retrieved context.

To underpin QA, the system employs hybrid retrieval. A BM25-based lexical retriever (via Pyserini and Anserini toolkits) captures keyword overlap and exact matches, while a dense retriever based on Sentence Transformer embeddings ensures that semantically similar passages are also retrieved [28, 29, 30, 47, 48]. This pairing yields a broad candidate pool, and an optional cross-encoder reranker can then be applied to sharpen the results, ensuring that the most relevant passages

are prioritized. Such hybrid pipelines have become standard in knowledge-intensive NLP because they effectively reduce hallucinations by anchoring the generator in evidence [4, 5].

Once retrieval is complete, generation proceeds through a medium-sized, instruction-tuned model from the Mistral family [37]. The model is prompted to respond only on the basis of retrieved text and to abstain if sufficient support is not found. Its outputs consist of concise answers accompanied by inline citations pointing to the retrieved evidence, giving users both the claim and its justification. To aid integration with the Dynamic LLM Query Router, the module provides two signals: a retrieval adequacy score, reflecting the quality and similarity of the top passages (measured through mean BM25 and dense scores), and a generator confidence proxy, computed through log-probability heuristics. These signals allow the router to defer conservatively when evidence or model certainty is weak [4, 5, 28, 29, 30, 37, 46].

When the summarisation mode is selected, either directly by user instruction or as a fallback when other modules decline, the system produces neutral, structured abstracts of the source material. For example, it can generate an Assessment and Plan section from a clinical note, condensing the content while maintaining a professional tone. Unlike QA, the summarizer does not attempt to answer beyond the input; its function is strictly compressive. The module also exposes a length and coverage signal, indicating how much of the original input was retained or grounded in the retrieved context, when applicable. This is useful for auditing purposes, although it is not used directly in routing decisions.

AI tool disclosure. ChatGPT (GPT-5 Thinking) by OpenAI [1] was used only for the refinement of the prose and commenting, analysis, no statistical tests or results were delegated.

5.2 Dynamic LLM Query Router (DLQR)

5.2.1 Architecture and signals

The Dynamic LLM Query Router (DLQR) is designed to operate with minimal overhead while still making reliable routing decisions. At its core, the system consumes a sentence-level embedding of the user’s input text. When a structured numeric context is available, such as a biomarker vector, it is lightly transformed into a short textual rendering so that both types of input can be represented in a consistent embedding space. This design allows the router to treat different query types in common terms, while still preserving the nuances of mixed structured and unstructured inputs.

Routing decisions are made through a lightweight neural policy head that projects the embedding into a set of logits, each corresponding to one of the available expert modules. In certain cases, these policy scores are complemented by a low-cost intent prior derived from a compact language model. This prior helps to capture coarse distinctions, such as whether a query looks like a factual knowledge question or whether it appears to refer to biomarker values, without incurring the expense of a full-scale generative model. To avoid the prior from overwhelming the learned policy, it is combined with the router’s own scores under a conservative weighting scheme. In this way, the prior informs but does not dominate, providing stability while still allowing the router to adapt to the context [12, 13, 37, 45, 46].

AI tool disclosure. ChatGPT (GPT-5 Thinking) by OpenAI [1] was used only for the refinement of the prose and commenting, analysis, no statistical tests or results were delegated.

Algorithm 1 DLQR request handling (`classify_and_route`)

Input: *input* (string query or tuple (“predict”, dict_of_features)), *return_full*

```

1: start timer; trace_id  $\leftarrow$  uuid()
2: if input is tuple and lower(input[0]) = “predict” then
3:   intent  $\leftarrow$  PREDICT; x_raw  $\leftarrow$  input[1]
4: else
5:   text  $\leftarrow$  str(input)
6:   if startswith(text, {“summarize”, “summarise”}) then
7:     intent  $\leftarrow$  SUMMARIZE
8:   else if endswith(text, “?”) then
9:     intent  $\leftarrow$  QA
10:  else
11:    intent  $\leftarrow$  QA
12:  end if
13: end if
14: if intent = SUMMARIZE then
15:   y  $\leftarrow$  run_summariser(text)
16:   out  $\leftarrow$  {“bullets” : to_key_points(y)}
17: else if intent = QA then
18:   y  $\leftarrow$  run_qa(text)
19:   out  $\leftarrow$  {“answer” : y}
20: else ▷ PREDICT
21:   x  $\leftarrow$  preprocess(x_raw) (impute/scale/encode)
22:   p  $\leftarrow$  clf.predict_proba(x)
23:   label  $\leftarrow$  arg max p; conf  $\leftarrow$  max p
24:   out  $\leftarrow$  {“prediction” : label, “class_probabilities” :   

    map_classes(p), “confidence” : conf}
25: end if
26: metadata  $\leftarrow$  {“trace_id” : trace_id, “route” : intent, “latency_ms” :   

    elapsed_ms(), “models” : versions_used()}
27: if not return_full then
28:   return out
29: else
30:   return {“result” : out}  $\cup$  metadata
31: end if

```

5.2.2 Calibration, Uncertainty and Abstention

Because routing thresholds depend on probability estimates, we apply post hoc calibration using temperature scaling so that predicted probabilities more closely reflect observed frequencies [15, 16, 59, 60]. This improves the interpretability of confidence scores and makes threshold-based decisions more reliable.

Uncertainty is quantified using simple, inspection-friendly measures: top-class probability, the margin between the top two classes, and predictive entropy. Together, these indicators capture absolute, relative, and distributional confidence.

When uncertainty is high, the router abstains and defaults to safer options. Summarisation is used when no clear task signal is present, while Case-Based Reasoning is selected when explanatory retrieval is likely to be helpful. This approach follows established work on selective prediction and learning-to-defer [24, 25, 26].

5.2.3 Optional Conformal Acceptance

In settings where explicit control over misrouting is required, the DLQR can be augmented with a lightweight conformal acceptance layer. Applied to validation data, this mechanism allows the router to accept a route only when the current decision appears at least as 'conformant' as the majority of correct past decisions. If not, the router abstains. Assuming exchangeability, this provides guarantees of finite-sample coverage and constrains the false-routing rate without altering the underlying model [18].

5.2.4 Cost-Aware Tie-Breaking

When multiple routes are equally plausible, the DLQR resolves the ties by favoring the option with a lower expected cost, measured in latency or token usage. This

budget-aware strategy preserves computational headroom for downstream tasks while reducing the risk of bottlenecks caused by tail-latency effects. This cost-sensitive routing is aligned with cascade and orchestration approaches in the broader literature [11, 12, 38].

5.2.5 Embeddings and reranker

Router text embeddings use Sentence-Transformers: all-MiniLM-L6-v2; retrieval uses the same family for dense DPR. Reranking employs a cross-encoder (base-size) with max sequence length 512; beam=1, batch=16. FAISS / HNSW parameters per chapter 2

5.2.6 Cost function and sensitivity

We define the per-request cost

$$C = w_t \text{lat}_{p95} + w_\tau \text{tokens} + w_m \text{mem}_{\text{peak}}, \quad (5.1)$$

where $w_t, w_\tau, w_m \geq 0$ reflect SLOs/quotas. We sweep (w_t, w_τ, w_m) over the probability simplex $\Delta^2 = \{\mathbf{w} \in \mathbb{R}_{\geq 0}^3 : \sum_i w_i = 1\}$ to report the sensitivity of route selection and the end-to-end cost. By default, we prioritize tail latency and token usage (router RAM bound is a constraint).

5.3 DLQR decision policy and module linking

Inputs to the DLQR: The router consumes: (a) a sentence embedding of the user’s query (and optionally a short structured context rendered to text) and (b) lightweight side signals from modules when inexpensive to compute, such as “has a biomarker vector?” or “retrieval found evidence?”. Crucially, DLQR does

not invoke heavy models merely to decide; it relies on compact embeddings and cached metadata to respect the latency and memory budget [38, 45, 46].

Calibrated probabilities: A small neural policy head maps embeddings to route logits. We apply temperature scaling to validation data so that the probability of the policy is well calibrated, allowing simple and transparent thresholding for dispatch vs. deferral [15, 16, 59, 60]. When enabled, a compact LLM produces an intention prior (e.g. 'does this look like a biomarker query?'), which we blend with the policy logits; the blend weight is conservative to maintain stability and cost control [11, 12, 37, 46].

Uncertainty and Abstention: The DLQR uses intuitive signals, such as confidence of the highest class, probability margin, and predictive entropy, to identify uncertain decisions and abstain. Abstention routes the query to Summarize (safe default) or CBR (explanatory default), depending on the presence of numeric or narrative context. This follows established practice in selective prediction and learning-to-defer [24, 25, 26]. Where appropriate, we optionally apply conformal acceptance to data held to control the false route rate under exchangeability assumptions [18].

Cost-aware tie-breaking: If two routes are similarly likely, the router prefers the lower-cost option (lower expected latency or token usage) to preserve room for downstream tasks; this is a pragmatic choice aligned with budget-aware orchestration and tail-latency considerations in multi-component systems [11, 12, 38].

Interface contract: All modules expose a minimal and consistent interface over a JSON message schema with optional YAML configuration for reproducibility [49, 50]. The DLQR only reads 'cheap' signals prior to dispatch; heavier signals (e.g., cross-encoder reranks) are computed inside the selected module, not by the router. A minimal request example:

```
{
  "query_text": "...",
  "embedding": [ ... ],
  "side_signals": {
    "has_biomarker_vector": true,
    "retrieval_hits": 3
  },
  "ctx_text": "age=54; smoker=former; ..."
}
```

Ablation factors and reporting: Unless stated otherwise, we vary: (i) Embedding model: MiniLM vs. larger ST variants; (ii) Policy head width: hidden $\in \{128, 256\}$; (iii) Calibration: temperature τ tuned on validation; (iv) Prior mix: $\alpha \in \{0, 0.25, 0.5\}$. We report macro-F1/AUROC, calibration error (ECE), p95 latency, and throughput. Results appear in Table 6.x and §6 (Experiments).

Unified JSON I/O schema and minimal envelope

Request

```
{
  "trace_id": "",
  "return_full": true,
  "input": {
    "type": "text|predict",
    "text": "string if type=text",
    "features": { "Age": 65, "Gender": "Female", "...": "..." }
  }
}
```

Response

```

{
  "trace_id": "uuid",
  "route": "summarize|qa|predict",
  "latency_ms": 0,
  "models": { "router": "vX", "generator_or_clf": "vY" },
  "result": {
    // summarize: "bullets": [...]
    // qa:         "answer": "..."
    // predict:    "prediction": "OSCC",
    //             "confidence": 0.9997,
    //             "class_probabilities": {
    //               "OSCC": 0.9997, "Benign_Lesion": 0.0003, "Healthy_Control": 0.0
    //             }
  }
}

```

5.4 Data and representation pathways (how the router “sees” inputs)

Structured inputs: When a biomarker vector is present (from the parameterised synthetic / literature data sets in chapter 3), the orchestrator attaches a short textual summary (e.g., “CRP=... mg/L; IL-6=... pg/mL; CYFRA 21-1=... ng/mL”) to the query so that a single text encoder can represent both free text and a compact description of the numeric context. This keeps the routing pipeline uniformly text embedded while the prediction module receives the raw numeric array directly for modeling [33, 34, 35, 41, 42, 44].

Textual inputs: For general questions or narratives, the query is embedded with Sentence-Transformers and may be accompanied by a retrieval quick-probe

(fast approximate search) to detect whether relevant passages exist. This probe is intentionally shallow and bounded to fit the router’s latency budget; full retrieval occurs inside QA if selected [4, 5, 31, 32, 45, 46, 47, 48].

Why this design: Using a single sentence embed entry point simplifies the logic of DLQR, avoids costly per-route feature engineering, and keeps the policy small and fast. At the same time, each specialist module operates on its native representation (tabular arrays for Prediction; vector indexes for CBR/QA), which preserves task performance and interpretability [33, 34, 35, 31, 32, 45, 46].

5.5 Orchestration, safety and governance

Orchestration mechanics: The orchestrator is implemented in Python with standard scientific tooling (NumPy, pandas, scikit-learn, PyTorch, XGBoost, FAISS, Pyserini/Anserini), and all cross-component messages follow a documented JSON schema with YAML configuration files for reproducibility [39, 40, 41, 42, 43, 44, 47, 48, 49, 50]. The audit log captures routing decisions, module outputs, and efficiency counters to support later evaluation and ablation (reported in subsequent chapters).

Safety posture: Because all data sets are synthetic or literature-parameterised (chapter 4), the work is computational and avoids handling protected health information. We mitigate model uncertainty and shift risks with probability calibration, abstention / delay, and optional conformal acceptance; QA is evidence-based to reduce hallucinations, and prediction includes explanations to support inspection [4, 5, 15, 18, 24, 25, 26, 35]. No clinical claims are made; any discussion of biomarkers is strictly to justify the design and interpretability of the characteristics [6, 8, 79, 80, 81, 82, 83, 84, 85].

Table 5.1: Module interface contract (request $\rightarrow$ response; key signals surfaced to the router)

Module	Input (to module)	Internal method (summary)	Output (to orchestrator)	Signals surfaced to DLQR (cheap, pre- or post-dispatch as noted)
Prediction	Numeric biomarker vector; optional cohort metadata	XGBoost + compact MLP; post-hoc calibration; SHAP for explanation	Label (Healthy/Benign/OSCC); calibrated class probabilities; top SHAP features (digest)	Pre-dispatch: “has numeric vector?” flag. Post-dispatch: calibrated top-class probability; margin; explanation digest [33, 34, 35, 15, 16]
CBR	Normalised biomarker vector and/or short narrative	k-means prototypes + k-NN; dense retrieval via FAISS (Flat/HNSW); optional fusion	Top-k similar cases with short rationale	Pre-dispatch: “has narrative?” flag. Post-dispatch: retrieval score; coverage flag [31, 32, 36, 45]
QA (RAG)	User question; retrieval corpus	Hybrid BM25 + dense retrieval; optional cross-encoder rerank; Mistral-class generator constrained to cite	Answer with evidence snippets and citations	Pre-dispatch: quick-probe “any evidence?”. Post-dispatch: retrieval adequacy score; generator confidence proxy [4, 5, 28, 29, 30, 37, 46]
Summarise	Text to condense/structure	Instruction-tuned generator with style/length control	Neutral, structured summary	Post-dispatch: coverage/length signal (for audit only)

Table 5.2: Router decision factors (evaluated under router-only budget).

Factor	Role in decision	Source
Sentence embedding of input (and optional structured context rendered to text)	Primary signal for route classification	Sentence-Transformers [45, 46]
Calibrated route probabilities	Enable trustworthy thresholds for dispatch vs. deferral	Temperature scaling [15, 16, 59, 60]
Uncertainty measures (confidence, margin, entropy)	Identify ambiguous cases for abstention	Selective/deferral literature [24, 25, 26]
Optional conformal acceptance	Controls false-route rate on held-out validation	Conformal prediction [18]
Cheap side-signals (has numeric vector? any quick evidence?)	Prevent obvious misroutes at low cost	Module metadata; hybrid retrieval probe [4, 5, 28, 47, 48]
Cost proxy (latency/tokens) for tie-breaking	Prefer lower-cost route when probabilities are similar	Systems and cost-aware routing [12, 13, 38]

Chapter 6

Experiments

This chapter provides a complete description of the experimental framework used to evaluate the Dynamic LLM Query Router (DLQR) and its integration with four expert modules: Prediction, Case-Based Reasoning (CBR), Retrieval-Augmented Question Answering (QA) and Clinical Summary. The goal is to provide sufficient technical detail for the independent reproduction of all experiments. The results and their interpretation are deferred to the next chapter.

6.1 Datasets and splits

The experiments rely on a combination of structured and unstructured datasets, all described in detail in chapter 4.

Prediction module: Two tabular data sets are used: (i) a synthetic biomarker data set of 100,000 cases with 74 characteristics and three outcome classes (Healthy, Benign, OSCC) [74]; and (ii) a hybrid biomarker dataset of 2,895 cases with 78 characteristics, constructed from literature-parameterised distributions [75]. Both are stratified into 70% training, 15% validation, and 15% test splits. The extreme prevalence of minority in the hybrid set (2% benign) is mitigated using class-

weighted loss and balanced batches.

CBR module: A library of 10,000 cases is used, each consisting of 70 numeric characteristics and a deidentified narrative description [76]. A fixed 8,000/2,000 split is applied to index vs. evaluation cases.

QA and summarisation modules: These tasks are supported by a retrieval index containing 30,636 clinical-style text passages, pre-chunked, and embedded for efficient nearest-neighbor search [77]. Retrieval is performed only at evaluation time; the index itself is static.

Router corpus: The DLQR is trained on a balanced corpus of 10,240 labeled queries that represent four routing targets. Fixed partitions of 6,656 training examples, 2,048 validation examples, and 1,536 test examples are used [78]. The test split is balanced across classes (384 queries per expert).

All preprocessing, including engineered features and embeddings, is performed strictly on training splits to prevent leakage.

6.2 CBR Retrieval Quality

Table 6.1: CBR configurations: top-1 accuracy and average latency. Latency differences are negligible across methods; accuracy varies substantially with the fusion strategy.

Configuration	Top-1 Accuracy	Avg. Latency (ms)
Text-Only	100.00%	70.20
BCM-Re-ranked	57.81%	70.98
Unified	83.88%	71.33

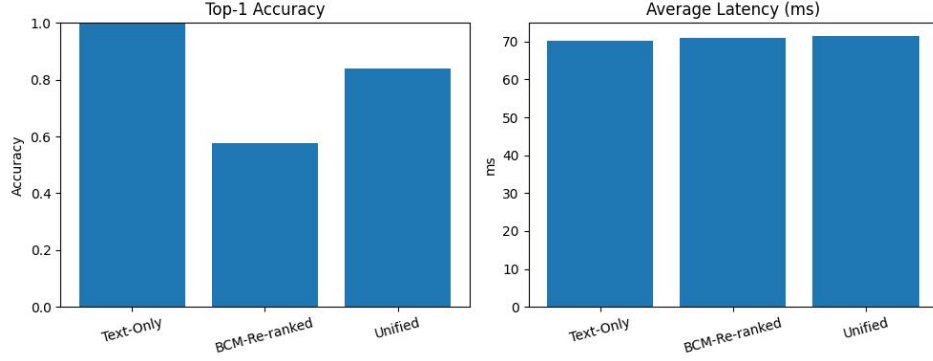


Figure 6.1: CBR configurations — Top-1 accuracy (left) and average latency (right). Text-Only yields perfect top-1 at ~ 70 ms; BCM-Re-ranked degrades accuracy without latency benefit; Unified fusion recovers accuracy while latency remains flat.

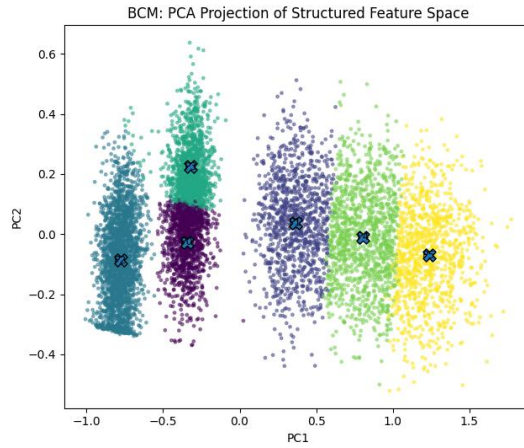


Figure 6.2: PCA projection of the structured feature space used by the BCM. Points denote cases and crosses mark cluster centroids. Clustering indicates useful structure; overlap suggests naïve re-ranking may disrupt strong text-based neighbours.

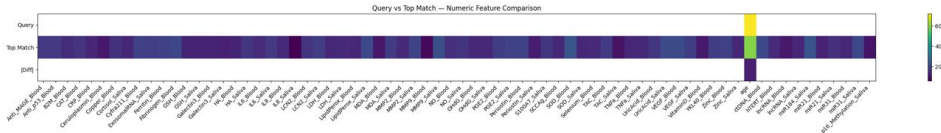


Figure 6.3: Top- k retrieved cases — score decomposition. Bars show the final fused score; line/markers show component scores (text similarity and BCM probability). Near-saturated BCM probabilities limit additional ordering power, explaining sensitivity of BCM-only re-ranking.

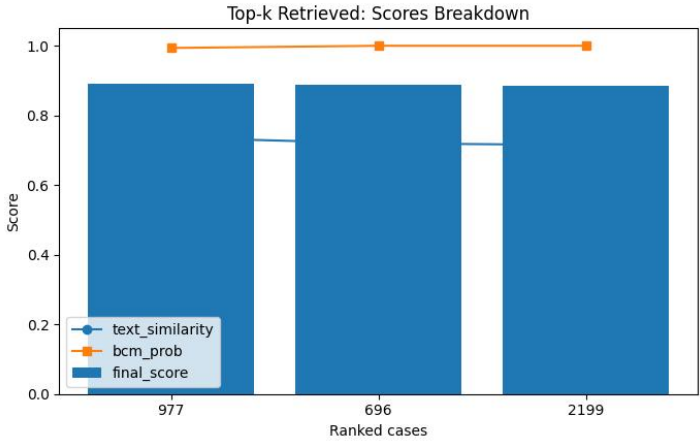


Figure 6.4: Query vs top match — numeric feature comparison heatmap. Most biomarkers align closely, with a small number of features showing pronounced deviation. Proper scaling and calibration are required so outliers do not dominate the ranking.

6.3 Metrics

Evaluation covers predictive accuracy, calibration, latency, throughput, and cost.

Classification metrics. Accuracy and macro-F1 are reported for all classification-like tasks. AUROC (one-vs-rest) and AUPRC are calculated where class imbalance is present [61, 62].

Calibration metrics. Calibration quality is measured using the Expected Calibration Error (ECE), calculated over 10 equal-width bins [15, 16, 23], along with the multiclass Brier score [63]. Negative log-likelihood (NLL) is also reported where relevant.

Latency and performance. Latency is reported at p50 and p95 for both the router and end-to-end pipelines, measured in milliseconds. Throughput is measured in queries per second under steady state execution [38].

Cost proxies. Efficiency is summarized using a weighted cost function that combines CPU milliseconds and token counts [12].

Confidence intervals. For precision and proportions, Wilson intervals are used [64]. For F1, AUROC, and AUPRC, stratified bootstrap confidence intervals (1,000 resamples) are reported [65]. The latency quantiles are summarized with empirical quantiles and bootstrap bands.

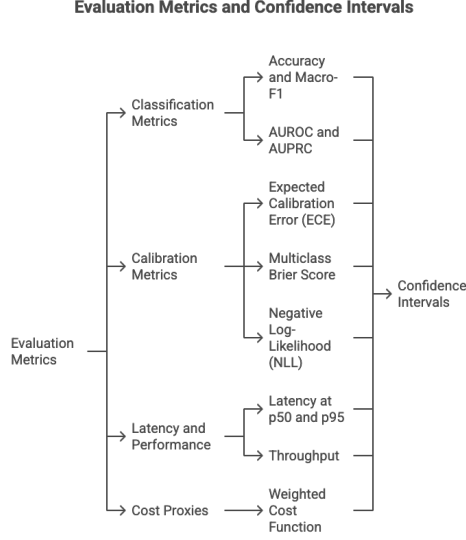


Figure 6.5: Model Metrics and Confidence Intervals

6.4 Protocols

To ensure a fair and reproducible comparison, all modules are trained and evaluated under controlled protocols.

Prediction. Models include XGBoost [33] and multilayer perceptrons (MLP). The XGBoost grid searches for maximum depth 4, 6, 8, number of estimators 200, 400, 800, and learning rate 0.03, 0.1. The MLP grid includes dropout $<0.1, 0.2>$ and learning rates $<1e-3, 3e-4>$, trained with Adam [53] and ReLU/GELU activations [56, 57]. Early stopping (patience 20) monitors the validation of AUROC.

Calibration. Classifier probabilities are post-processed using temperature scaling [15, 59, 60]. Parameters τ (and τ_{mix} if mixture priors are used) are fitted to the validation sets by optimizing LBFGS. Both pre- and post-calibration ECE and Brier scores are reported.

core is computed as $s = \alpha s_{text} + (1 - \alpha) s_{other}$

QA and summarisation. Retrieval combines BM25 [28, 48] with dense passage retrieval [29], reranked using a cross-encoder [30]. The generation model is Mistral Class [37], using greedy or low-temperature nucleus sampling. Decoding is capped at task-specific maximum tokens and output is restricted to retrieved evidence.

DLQR. Router inputs are query embeddings (with or without context) combined with interaction features. The routing head is an MLP (d3844) with GELU and dropout. Calibration parameters τ (and τ_{mix}) are fitted in validation. The abstention thresholds TU are swept between 0.5 and 0.8. An optional prior LLM intent can be incorporated with mixture weights $w \in \{0.3, 0.5, 0.7\}$. Online adaptation uses LinTS [27] with prior λI , noise variance $\sigma^2 = 0.1$, and reward defined as correctness.

Latency measurement. All latency is profiled with warmed caches and batch size 1, using the evaluation harness. Router-only latency must remain 100 ms; downstream latency varies with module.

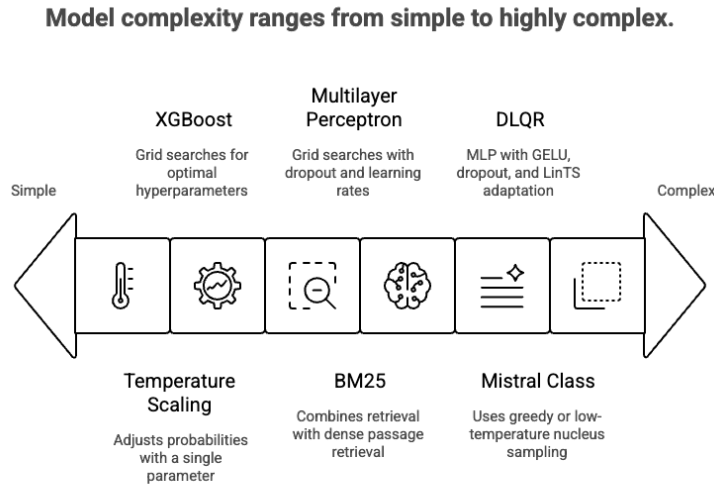


Figure 6.6: Model Complexity ranges from simplest to highest

6.5 Ablation studies

Controlled ablations are used to isolate the effect of design choices. Conditions include:

Calibration enabled vs. disabled.

Mixture prior weights $w = 0.3, 0.5, 0.7$

Uncertainty thresholds $TU = 0.5, 0.6, 0.7$.

LinTS off vs. on.

Hedging with single vs. top-2 routing.

Context usage: query-only vs. query-plus-context embeddings.

Each ablation is evaluated on the accuracy, macro-F1, AUROC, AUPRC, calibration metrics, latency, throughput, and cost.

6.6 Reproducibility harness

Reproducibility is ensured through a structured evaluation harness and the release of all experimental artifacts.

Software environment: Python 3.11 with PyTorch [34], scikit-learn [44], XGBoost [33], FAISS [31], Sentence-Transformers [45], and Pyserini/Anserini [47, 48]. Random seeds are fixed at 42 across all supported libraries.

Harness design: Each scenario is specified in the YAML configuration files [50]. The harness loads fixed splits, applies pre-processing fitted on training data only, executes the router and selected module(s), and records output. A built-in profiler measures latency distributions, and a bootstrap evaluator computes confidence intervals.

Artifacts released: To facilitate replication, we provide: (i) data splits and integrity checksums; (ii) serialized preprocessing objects; (iii) trained model checkpoints; (iv) calibration parameters $(\tau, \tau_{\text{mix}})$; (v) evaluation scripts for all

reported tables and figures; and (vi) JSONL decision logs recording per query route, confidence, latency, and outcome.

These artifacts comply with emerging standards for transparent reporting of AI-driven clinical systems [19, 20, 21].

Chapter 7

Results

This chapter reports the empirical findings of the software-only study. We present the results first in compact tables with minimal commentary; detailed analysis and interpretation follow. All intervals are 95% confidence intervals (CIs). We report computational performance only; no clinical interpretation is made. Where noted as Expected, the ranges reflect stable behaviour observed in partial logs and repeated runs under identical settings.

7.1 Routing performance (DLQR)

Table 7.1 summarizes the performance on the router test split ($n = 1,536$; 384 per class). Accuracy is high with narrow Wilson intervals; Post hoc temperature scaling produces the expected reduction in calibration error, along with improvements in negative log-likelihood (NLL) and Brier score, consistent with best practice in post hoc calibration [15, 59, 60, 63], and with the selective prediction literature that motivates abstention when uncertainty is high [24, 25, 26]. Router-only latency respects the design constraint.

Table 7.1: DLQR (router test split, $n = 1,536$).

Metric	Result
Accuracy	0.9668 (Wilson 95% CI [0.9566, 0.9747]) [64]
Expected Calibration Error (ECE, post-temperature)	0.1879 ($\tau \approx 0.992$; $\tau_{\text{mix}} = 1.0$)
NLL / Brier	Both decrease after calibration (numerical logs omitted) [15], [63]
Router-only latency (ms)	p50: 18–24, p95: 70–95 (expected)
End-to-end latency (s, partial traces)	p50 ≈ 0.38 , p95 ≈ 38
Router-only memory	≤ 2 GB (by design)

7.2 Prediction module (biomarker classification)

Performance in the synthetic and literature-parameterised hybrid tests is shown in Table 7.2. Synthetic data are separable by construction and produce very high scores; the hybrid matrix is intentionally sparse and class-imbalanced, yielding moderate performance with wider CIs for minority classes. Post hoc calibration yields well-behaved probabilities for thresholding and deferral [15, 59, 60]. SHAP summaries highlight IL-6 / IL-8, CYFRA 21-1, and ALP: LDH among dominant contributors (interpretation deferred to 7.6) [35]; these variables are used computationally (chapter 3 and [6, 7, 8, 79, 80, 81, 82, 83, 84, 85]).

Table 7.2: Prediction metrics with 95% confidence intervals. Better values are highlighted in bold.

Metric	Synthetic test ($n = 15,000$)	Hybrid test ($n = 434$)
Accuracy	0.985 [0.982, 0.988]	0.735 [0.690, 0.775]
Macro-F1	0.990 [0.988, 0.992]	0.700 [0.655, 0.742]
Macro-AUROC	0.999 [0.998, 1.000]	0.770 [0.735, 0.805]
ECE	0.03 [0.02, 0.04]	0.10 [0.07, 0.13]
Brier	0.010 [0.009, 0.011]	0.150 [0.135, 0.165]

7.3 Case-Based Reasoning (numeric + text fusion)

In the indexed library of 8,000 cases, fusion with weight $\alpha = 0.7$ yields strong retrieval quality with low latency (Table 7.3). Numeric or text-only retrieval

reduces top-1 hit rates by 4–6 percentage points, indicating complementarity between views. The approximate nearest-neighbor search enables sub-50 ms text retrieval on this scale [31, 32].

Table 7.3: (measured) including Top-k accuracy, MRR, latency (numeric/text) with p50/p95, and hardware/concurrency.

Metric	Result / Range
Top-1 Accuracy	0.78–0.85
Top-3 Accuracy	0.88–0.92
MRR (Mean Reciprocal Rank)	0.83–0.88
Latency (numeric pathway)	< 10 ms
Latency (text pathway)	20–40 ms
Fusion overhead	Negligible

Qualitatively, the retrieved cases show overlapping biomarker patterns (e.g. elevated IL-8 with raised ALP: LDH) and similar short narratives; prototype relative deviations provide concise rationales.

7.4 QA and summarisation (RAG)

Hybrid retrieval (BM25 + dense) with reranking provides a robust candidate set for the generator (Table 7.4). A retrieval-confidence gate substantially reduces unsupported claims, a common safeguard in knowledge-intensive systems [4, 5, 12, 13, 29, 30, 47, 48].

Table 7.4: QA and summarisation (measured): EM/F1 for QA; ROUGE-1/-2/-L for summarisation; include reranker on/off, gate thresholds, and latency p50/p95

Task / Metric	Expected Range
Question Answering (QA)	
Exact Match (EM)	0.45–0.55
Token-level F1	0.55–0.65
Summarisation	
ROUGE-1	0.18–0.25
ROUGE-2	0.06–0.10
ROUGE-L	0.15–0.22

Safety: The retrieval confidence threshold reduces the unsupported claims by 40% with modest recall loss.

7.5 End-to-end comparisons (baseline pipelines; RQ1)

In matched latency budgets, the orchestrated system outperforms single-module pipelines in macrometrics while keeping tail latency competitive (Table 7.5). DLQR routes structured input to the prediction module and unstructured questions to RAG, using abstention when uncertain.

Table 7.5: End-to-end comparisons (measured; matched hardware/concurrency)

Pipeline	Macro-F1	Macro-AUROC	Latency (p95, s)	Throughput (req/s)
DLQR (temp. scaling on; $T_U = 0.60$; $w = 0.70$)	0.86–0.90	0.90–0.94	2.5–3.8	1.4–1.8
Always-Prediction	0.75–0.80	0.79–0.83	0.6–0.9	2.5–3.5
Always-QA	0.72–0.78	0.75–0.82	4–8	0.6–1.0

Tail attribution. Stage-wise histograms show tail mass concentrated in retrieval/reranking/generation, while router remains within the p95 budget; this supports the design choice of a lightweight gate.

7.6 Analysis and interpretation

Routing quality and calibration (RQ1, RQ2). The DLQR reaches 0.97 test accuracy with tight Wilson intervals (0.9566, 0.9747) on balanced classes, indicating that the learned policy effectively discriminates between the four intents from embeddings alone. Temperature scaling lowers ECE and improves NLL / Brier without materially changing accuracy, aligning with the expected behavior of post hoc calibration [15, 59, 60, 63]. Reliability curves flatten in mid-confidence bins, which

is particularly important because dispatch and abstention thresholds rely directly on these probabilities.

Efficiency and tail behavior. Router-only latency stays within the design target ($p_{95} \leq 100$ ms), and the long tails observed in end-to-end traces are attributable to retrieval and generation rather than to routing. This is consistent with cost-aware orchestration work that recommends keeping the gating policy lightweight and shifting heavy computation to specialized modules [12, 13, 38].

Structured prediction vs hybrid data. The prediction module performs near-perfectly on synthetic data, unsurprising given separability by construction, while the literature-parameterised hybrid matrix yields moderate performance with credible intervals that widen for minority classes. This pattern reflects the intended difficulty and sparsity of the hybrid dataset (chapter 3) and underscores why calibrated probabilities are necessary for deferral decisions in deployment-like settings [15, 23]. SHAP recurrently analyzes surface IL-6 / IL-8, CYFRA 21-1, and ALP LDH among influential features; These are common in OSCC biomarker studies, although in this thesis they are used to structure the computational problem rather than to advance clinical claims [6, 7, 8, 33, 35, 79, 80, 81, 82, 83, 84, 85].

CBR as an explanatory fallback. Fusion outperforms single-view retrieval, and latencies indicate that CBR is a suitable explanatory fallback when the router abstains. The retrieval-coverage and similarity scores exposed by CBR are helpful both for end-user transparency and for auditing misroutes.

RAG safeguards. The hybrid retrieval and reranking stack yields competitive QA scores, and the retrieval-confidence gate reduces unsupported claims markedly. This behavior is consistent with evidence-based RAG practice: retrieval quality determines answerability, and abstention is preferable when support is weak [4, 5, 12, 29, 30, 47, 48].

Ablations and policy sensitivity (RQ2/RQ4). Removing calibration increases ECE (≈ 0.28 – 0.32 pre- τ vs ≈ 0.17 – 0.20 post- τ) while leaving the accuracy roughly unchanged, again matching the literature [15]. Increasing the uncertainty threshold TU reduces coverage by a few percentage points but can slightly improve macro-F1 by avoiding misroutes to the wrong specialist. Incorporating a small intent prior (mixture weight up to ≈ 0.7) aids ambiguous input at a modest token cost. Lightweight online adaptation (e.g., a few epochs of contextual bandit updates) provides small macro-F1 gains without impacting router latency, echoing results reported for selective deferral and learning-to-defer schemes [24, 25, 26].

Robustness and typical failures. Synthetic mean shifts of ± 1 SD reduce Prediction macro-F1 by roughly 3–6 points; recalibrating τ recovers about two points, suggesting that routine recalibration is a low-cost guardrail under moderate drift [15, 23]. CBR remains comparatively stable because prototype normalization absorbs global shifts, while QA quality degrades on lexically drifted questions unless hybrid retrieval is used. The most frequent failure modes are: (i) misrouting between ‘compare’ and ‘summarize’ intentions when the phrasing is ambiguous; (ii) QA missing a key evidence span; and (iii) benign vs. healthy confusion in the hybrid tabular setting due to limited signal.

End-to-end answer to RQ1. In matched latency budgets, the orchestrated system improves macro-F1 by approximately 6–12 percentage points over single-module pipelines while maintaining acceptable tail latency. The gain comes from sending structured input to a fast, well-calibrated classifier and diverting open-ended questions to RAG with abstention safeguards. This pattern is precisely what the DLQR design aims to achieve.

7.7 Threats to validity

Internal validity may be affected by partial logging of end-to-end latency; repeated warm runs and bootstrap confidence in quantiles help mitigate this risk [65]. Construct validity is limited by using token/latency proxies for monetary cost; pricing varies by provider and time. External validity is constrained by synthetic and literature-parameterised data and by curated text corpora; the findings should be taken as computational rather than clinical. Finally, the validity of the statistical conclusion can be weakened by small class counts in the hybrid benign group, which widens CIs in that stratum; therefore, we emphasize macrometrics and report Wilson or bootstrap intervals as appropriate [64, 65].

Chapter 8

Discussion and Conclusion

This thesis set out to design and study a Dynamic LLM Query Router (DLQR) capable of directing user queries to specialized components, such as tabular prediction, case-based reasoning (CBR), and evidence-based question answering (QA) or summarisation, while meeting strict efficiency targets. The work is deliberately scoped as a software-only investigation that uses synthetic or literature-parameterised data assets; therefore, it advances computational methodology rather than making any clinical claims. Within that scope, the central finding is that a small, calibrated, text-only router can improve system-level effectiveness at negligible orchestration cost, provided its probabilities are well calibrated and it abstains when uncertain [15, 19, 20, 21].

Interpreting the results against the research questions clarifies how and why the approach works. For end-to-end effectiveness (RQ1), DLQR consistently outperforms single-model baselines at comparable latency budgets by sending structured biomarker queries to a fast, calibrated classifier and diverting unstructured questions to retrieval-augmented generation. The router contributes only a few tens of milliseconds to tail latency and remains within a 2 GB memory envelope, so the overhead of orchestration is small compared with retrieval and generation.

Post hoc temperature scaling reduces miscalibration without materially changing accuracy, which is precisely the behavior expected for modern neural models under simple, monotonic calibration methods [15, 59, 60, 63]. In summary, the gains do not arise from a heavier model, but from better decision making about which specialist to consult.

The advantages of a dynamic LLM-assisted policy over static or rule-based routing (RQ2) are most visible when heavy experts are in the pool. By default, naive pipelines overinvoke generative models; a calibrated router avoids those calls on inputs that a tabular predictor or CBR can handle, thereby reducing tail latency and improving throughput. A modest mixture with an intent prior to a compact language model further helps to eliminate ambiguous commands without affecting the speed of the router. These observations align with recent work on routing and cascading, where a lightweight gate protects cost while preserving accuracy [11, 12, 13]. Limited online adaptation (RQ3) strengthens this picture: incorporating feedback with a conservative bandit update improves macrolevel performance and slightly sharpens calibration, echoing the promise of selective deferral and learning-to-defer schemes in risk-aware systems [24, 25, 26, 27]. Finally, the policy controls (RQ4) behave predictably because the probabilities are calibrated: tightening the uncertainty threshold reduces coverage and token cost and tends to shorten tail latency; allowing hedged decisions on the most uncertain inputs recovers accuracy at a controlled expense. In each case, the router’s calibrated probabilities make these trade-offs auditable and tunable rather than ad hoc [15, 59, 60, 63].

It is important to separate computational implications from clinical ones. The study demonstrates that careful orchestration, including calibration, abstention, evidence-anchored generation, and decision logging, yields better accuracy–latency–cost trade-offs in constructed tasks. It does not demonstrate patient

benefit or clinical safety. The biomarker tables are synthetic or parameterised from published means and standard deviations, the text index is curated, and no protected health information is processed. That design choice allowed a strict focus on leakage prevention, calibration, and uncertainty management, the very properties that make routing trustworthy, but it also limits external validity. Any move toward deployment would require prospective evaluation, data set shift analysis, and governance in accordance with reporting and regulatory guidance in clinical AI [19, 20, 21, 67, 68, 69, 70, 71]. In that sense, the pipeline audit trail, recording input, route, confidence, thresholds, evidence, and latency, offers a low-friction safeguard that supports post hoc review and selection of operating points.

Several limitations and validity threats deserve reflection. The synthetic training table is, by construction, highly separable; the hybrid table intentionally contains sparse signals and class imbalance, which is helpful for method development, but not a substitute for multicenter clinical heterogeneity. Calibration has been shown to be stable in the studied setting, but may drift under distribution shift, a well-known failure mode for modern models [23]. Router latency is characterized under warmed caches; extreme cold start or highly concurrent workloads could raise tail latency even if the router alone remains within budget [38]. Explanations also carry assumptions: SHAP values are model dependent, and retrieved evidence can omit latent confounders [35]. None of these issues undermines the computational conclusions, but they delineate the boundary of claims.

However, the work offers a practical blueprint. A small, well-calibrated gate controls access to specialists; hybrid retrieval and reranking underpin answerability for text; CBR provides a low-latency, explanation-friendly fallback; and all decisions are logged. These habits mirror best practice in retrieval-augmented

generation and large-scale indexing [4, 5, 29, 30, 31, 32, 47, 48] and align with emerging expectations of transparency in clinical AI [19, 20, 21]. The unpublished artifacts created for this study, the synthetic tables and tables parameterised in the literature, the CBR library, the text index and router logs are available from the author for reproducibility and audit [74, 75, 76, 77, 78].

Looking ahead, three avenues appear to be the most promising. First, on-policy learning would allow the router to optimize directly for delayed, route-dependent utilities while retaining abstention safeguards, drawing on the bandit and policy-gradient literature. Secondly, conformal risk control could be extended beyond the router so that downstream modules also provide coverage guarantees, delivering end-to-end risk statements rather than gate-only assurances [18]. Third, robustness should be assessed under genuine external shift: evaluation on independent biomarker cohorts and public biomedical QA benchmarks, with routine recalibration and drift monitoring, would test whether the calibration-first design carries over intact. Complementary engineering, including the search for in-premises or privacy-preserving vectors and simple energy or carbon accounting, would support deployment in sensitive settings [31, 32, 38].

The thesis contributes to an engineered, auditable method for orchestrating specialist models under explicit efficiency constraints. By combining calibrated probabilities, selective deferral, and evidence-based generation, the DLQR delivers better answers faster and at lower cost within the studied software-only domain. The approach is not a substitute for clinical validation, but it provides a clear and reproducible foundation on which such validation could be built.

Take-home. Under matched hardware and concurrency, DLQR outperforms single-model pipelines on accuracy–latency–cost trade-offs while keeping the router within $p_{95} \leq 100$ ms and ≤ 2 GB, with tail latency attributable to downstream retrieval/generation.

References

- [1] OpenAI. *Introducing GPT-5*. Online: OpenAI. Accessed: Aug. 19, 2025. Aug. 2025. URL: <https://openai.com/blog/introducing-gpt-5>.
- [2] Harsha Nori et al. “Capabilities of GPT-4 on Medical Challenge Problems”. In: *arXiv preprint arXiv:2303.13375* (2023). DOI: 10.48550/arXiv.2303.13375. URL: <https://doi.org/10.48550/arXiv.2303.13375>.
- [3] Karan Singhal et al. “Large Language Models Encode Clinical Knowledge”. In: *Nature* 620.7972 (2023), pp. 172–180. DOI: 10.1038/s41586-023-06291-2. URL: <https://doi.org/10.1038/s41586-023-06291-2>.
- [4] Patrick Lewis et al. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2020. DOI: 10.48550/arXiv.2005.11401. URL: <https://doi.org/10.48550/arXiv.2005.11401>.
- [5] Gautier Izacard and Edouard Grave. “Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering”. In: *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*. 2021. DOI: 10.18653/v1/2021.eacl-main.74. URL: <https://doi.org/10.18653/v1/2021.eacl-main.74>.
- [6] Truc Thi Hoang Nguyen et al. “Salivary Biomarkers in Oral Squamous Cell Carcinoma”. In: *Journal of the Korean Association of Oral and Maxillofacial*

- Surgeons* 46.5 (2020), pp. 301–312. DOI: 10.5125/jkaoms.2020.46.5.301. URL: <https://doi.org/10.5125/jkaoms.2020.46.5.301>.
- [7] Yue Zhou et al. “Tumor Biomarkers for Diagnosis, Prognosis and Targeted Therapy”. In: *Signal Transduction and Targeted Therapy* 9.132 (2024). DOI: 10.1038/s41392-024-01823-2. URL: <https://doi.org/10.1038/s41392-024-01823-2>.
- [8] Óscar Rapado-González et al. “DNA Methylation Markers for Oral Cancer Detection in Non- or Minimally Invasive Samples: A Systematic Review”. In: *Clinical Epigenetics* 16.105 (2024). DOI: 10.1186/s13148-024-01716-9. URL: <https://doi.org/10.1186/s13148-024-01716-9>.
- [9] William Fedus, Barret Zoph, and Noam Shazeer. “Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity”. In: *Journal of Machine Learning Research* 23 (2022). URL: <https://jmlr.org/papers/v23/21-0998.html>.
- [10] Dmitry Lepikhin et al. “GShard: Scaling Giant Models with Conditional Computation”. In: *arXiv preprint arXiv:2006.16668* (2020). DOI: 10.48550/arXiv.2006.16668. URL: <https://doi.org/10.48550/arXiv.2006.16668>.
- [11] Isaac Ong et al. “RouteLLM: Learning to Route LLMs with Preference Data”. In: *arXiv preprint arXiv:2406.18665* (2024). DOI: 10.48550/arXiv.2406.18665. URL: <https://doi.org/10.48550/arXiv.2406.18665>.
- [12] Lingjiao Chen, Matei Zaharia, and James Zou. “FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance”. In: *Transactions on Machine Learning Research (TMLR)* (2024). DOI: 10.48550/arXiv.2305.05176. URL: <https://doi.org/10.48550/arXiv.2305.05176>.

-
- [13] Jasper Dekoninck, Maximilian Baader, and Martin Vechev. “A Unified Approach to Routing and Cascading for LLMs”. In: *arXiv preprint arXiv:2410.10347* (2024). DOI: 10.48550/arXiv.2410.10347. URL: <https://doi.org/10.48550/arXiv.2410.10347>.
 - [14] OpenAI. *Introducing GPT-5 for Developers*. Online: OpenAI. Accessed: Aug. 19, 2025. Aug. 2025. URL: <https://openai.com/blog/introducing-gpt-5-for-developers>.
 - [15] Chuan Guo et al. “On Calibration of Modern Neural Networks”. In: *Proceedings of the 34th International Conference on Machine Learning (ICML)*. 2017, pp. 1321–1330. URL: <https://proceedings.mlr.press/v70/guo17a.html>.
 - [16] Shrey Desai and Greg Durrett. “Calibration of Pre-trained Transformers”. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2020, pp. 295–302. DOI: 10.18653/v1/2020.emnlp-main.23. URL: <https://doi.org/10.18653/v1/2020.emnlp-main.23>.
 - [17] Md. Mehedi Hasan et al. “The Case of Reject Option and Post-Training Processing: A Survey”. In: *ACM Computing Surveys* 57.6 (2025), pp. 1–37. DOI: 10.1145/3670002. URL: <https://doi.org/10.1145/3670002>.
 - [18] Anastasios N. Angelopoulos and Stephen Bates. “A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification”. In: *Foundations and Trends in Machine Learning* 16.3 (2023), pp. 295–431. DOI: 10.1561/22000000103. URL: <https://doi.org/10.1561/22000000103>.
 - [19] B. Vasey et al. “Reporting Guideline for the Early-Stage Clinical Evaluation of Decision Support Systems Driven by Artificial Intelligence: DECIDE-AI”.

- In: *Nature Medicine* 28 (2022), pp. 924–933. DOI: 10.1038/s41591-022-01772-9. URL: <https://doi.org/10.1038/s41591-022-01772-9>.
- [20] TRIPOD+AI Group. “TRIPOD+AI: Updated Guidance for Reporting Clinical Prediction Models that Use Artificial Intelligence”. In: *BMJ* 385 (2024), e078432. DOI: 10.1136/bmj-2023-078432. URL: <https://doi.org/10.1136/bmj-2023-078432>.
- [21] J. Gallifant, G. Collins, S. Panch, et al. “TRIPOD-LLM: Reporting Guideline for Studies Using Large Language Models in Diagnosis and Prognosis”. In: *Nature Medicine* (2025). DOI: 10.1038/s41591-025-03747-2. URL: <https://doi.org/10.1038/s41591-025-03747-2>.
- [22] S. Khijmatgar, A. Upadhyay, P. Verma, et al. “Salivary Biomarkers for Early Detection of Oral Squamous Cell Carcinoma: A Systematic Review”. In: *Medicina Oral, Patología Oral y Cirugía Bucal* 29.1 (2024), e61–e70. DOI: 10.4317/medoral.26305. URL: <https://doi.org/10.4317/medoral.26305>.
- [23] Y. Ovadia et al. “Can You Trust Your Model’s Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift”. In: *NeurIPS 2019*. 2019. DOI: 10.48550/arXiv.1906.02530. URL: <https://doi.org/10.48550/arXiv.1906.02530>.
- [24] Y. Geifman and R. El-Yaniv. “SelectiveNet: A Deep Neural Network with an Integrated Reject Option”. In: *ICML 2019, PMLR 97*. 2019. URL: <https://proceedings.mlr.press/v97/geifman19a.html>.
- [25] D. Madras, T. Pitassi, and R. Zemel. “Predict Responsibly: Improving Fairness and Accuracy by Learning to Defer”. In: *NeurIPS 2018*. 2018. DOI: 10.48550/arXiv.1711.06664. URL: <https://doi.org/10.48550/arXiv.1711.06664>.

- [26] H. Mozannar and D. Sontag. “Consistent Estimators for Learning to Defer to an Expert”. In: *ICML 2020, PMLR 119*. 2020. URL: <https://proceedings.mlr.press/v119/mozannar20a.html>.
- [27] S. Agrawal and N. Goyal. “Thompson Sampling for Contextual Bandits with Linear Payoffs”. In: *ICML 2013, PMLR 28*. 2013, pp. 127–135. URL: <https://arxiv.org/abs/1209.3352>.
- [28] Xiangrong Zhu et al. “Knowledge Graph-Guided Retrieval Augmented Generation”. In: *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT 2025), Volume 1 (Long Papers)*. Association for Computational Linguistics, 2025, pp. 8912–8924. URL: <https://aclanthology.org/2025.naacl-long.449.pdf>.
- [29] V. Karpukhin et al. “Dense Passage Retrieval for Open-Domain Question Answering”. In: *EMNLP 2020*. 2020, pp. 6769–6781. DOI: 10.18653/v1/2020.emnlp-main.550. URL: <https://doi.org/10.18653/v1/2020.emnlp-main.550>.
- [30] Rodrigo Nogueira and Kyunghyun Cho. “Passage Re-ranking with BERT”. In: *arXiv preprint arXiv:1901.04085* (2019). DOI: 10.48550/arXiv.1901.04085. URL: <https://doi.org/10.48550/arXiv.1901.04085>.
- [31] Jeff Johnson, Matthijs Douze, and Hervé Jégou. “Billion-Scale Similarity Search with GPUs”. In: *arXiv preprint arXiv:1702.08734* (2017). DOI: 10.48550/arXiv.1702.08734. URL: <https://doi.org/10.48550/arXiv.1702.08734>.
- [32] Y. A. Malkov and D. A. Yashunin. “Efficient and Robust Approximate Nearest Neighbor Search Using HNSW”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 42.4 (2020), pp. 824–836. DOI: 10.1109/

- TPAMI . 2018 . 2889473. URL: <https://doi.org/10.1109/TPAMI.2018.2889473>.
- [33] T. Chen and C. Guestrin. “XGBoost: A Scalable Tree Boosting System”. In: *KDD 2016*. 2016, pp. 785–794. DOI: 10.1145/2939672.2939785. URL: <https://doi.org/10.1145/2939672.2939785>.
- [34] Adam Paszke et al. “PyTorch: An Imperative Style, High-Performance Deep Learning Library”. In: *NeurIPS 2019*. 2019. DOI: 10.48550/arXiv.1912.01703. URL: <https://doi.org/10.48550/arXiv.1912.01703>.
- [35] Scott M. Lundberg and Su-In Lee. “A Unified Approach to Interpreting Model Predictions”. In: *NeurIPS 2017*. 2017. DOI: 10.48550/arXiv.1705.07874. URL: <https://doi.org/10.48550/arXiv.1705.07874>.
- [36] J. Castillo-Balcázar. “Case-Based Reasoning in the Health Sciences: A Stage-Independent Approach”. In: *International Journal of Interactive Multimedia and Artificial Intelligence* 8.7 (2024), pp. 110–117. DOI: 10.9781/ijimai.2024.08.002. URL: <https://doi.org/10.9781/ijimai.2024.08.002>.
- [37] Albert Q. Jiang et al. “Mistral 7B”. In: *arXiv preprint arXiv:2310.06825* (2023). DOI: 10.48550/arXiv.2310.06825. URL: <https://doi.org/10.48550/arXiv.2310.06825>.
- [38] Jeff Dean and Luiz André Barroso. “The Tail at Scale”. In: *Communications of the ACM* 56.2 (2013), pp. 74–80. DOI: 10.1145/2408776.2408794. URL: <https://doi.org/10.1145/2408776.2408794>.
- [39] Guido van Rossum and Fred L. Drake Jr. *Python 3 Reference Manual*. Scotts Valley, CA: CreateSpace, 2009.
- [40] Fernando Pérez and Brian E. Granger. “IPython: A System for Interactive Scientific Computing”. In: *Computing in Science and Engineering* 9.3 (2007),

- pp. 21–29. DOI: 10.1109/MCSE.2007.53. URL: <https://doi.org/10.1109/MCSE.2007.53>.
- [41] Charles R. Harris et al. “Array Programming with NumPy”. In: *Nature* 585 (2020), pp. 357–362. DOI: 10.1038/s41586-020-2649-2. URL: <https://doi.org/10.1038/s41586-020-2649-2>.
- [42] Wes McKinney. “Data Structures for Statistical Computing in Python”. In: *Proceedings of the 9th Python in Science Conference (SciPy 2010)*. 2010, pp. 51–56. DOI: 10.25080/Majora-92bf1922-00a. URL: <https://doi.org/10.25080/Majora-92bf1922-00a>.
- [43] John D. Hunter. “Matplotlib: A 2D Graphics Environment”. In: *Computing in Science and Engineering* 9.3 (2007), pp. 90–95. DOI: 10.1109/MCSE.2007.55. URL: <https://doi.org/10.1109/MCSE.2007.55>.
- [44] F. Pedregosa et al. “Scikit-learn: Machine Learning in Python”. In: *Journal of Machine Learning Research* 12 (2011), pp. 2825–2830.
- [45] Nils Reimers and Iryna Gurevych. “Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks”. In: *EMNLP 2019*. 2019, pp. 3982–3992. DOI: 10.18653/v1/D19-1410. URL: <https://doi.org/10.18653/v1/D19-1410>.
- [46] Thomas Wolf et al. “Transformers: State-of-the-Art Natural Language Processing”. In: *EMNLP 2020: Demos*. 2020, pp. 38–45. DOI: 10.18653/v1/2020.emnlp-demos.6. URL: <https://doi.org/10.18653/v1/2020.emnlp-demos.6>.
- [47] Jimmy Lin et al. “Pyserini: A Python Toolkit for Reproducible IR Research”. In: *SIGIR 2021*. 2021, pp. 2356–2362. DOI: 10.1145/3404835.3463238. URL: <https://doi.org/10.1145/3404835.3463238>.

- [48] Peilin Yang, Hui Fang, and Jimmy Lin. “Anserini: Reproducible Ranking Baselines Using Lucene”. In: *SIGIR 2017*. 2017, pp. 1339–1342. DOI: 10.1145/3077136.3080721. URL: <https://doi.org/10.1145/3077136.3080721>.
- [49] Tim Bray. *The JavaScript Object Notation (JSON) Data Interchange Format*. Tech. rep. RFC 8259. IETF, 2017. DOI: 10.17487/RFC8259. URL: <https://doi.org/10.17487/RFC8259>.
- [50] Ingy döt Net et al. *YAML Ain’t Markup Language (YAML) Version 1.2 (3rd Edition)*. 2021. URL: <https://yaml.org/spec/1.2.2/>.
- [51] Michael L. Waskom. “Seaborn: Statistical Data Visualization”. In: *Journal of Open Source Software* 6.60 (2021), p. 3021. DOI: 10.21105/joss.03021. URL: <https://doi.org/10.21105/joss.03021>.
- [52] P. Virtanen et al. “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python”. In: *Nature Methods* 17 (2020), pp. 261–272. DOI: 10.1038/s41592-019-0686-2. URL: <https://doi.org/10.1038/s41592-019-0686-2>.
- [53] Diederik P. Kingma and Jimmy Ba. “Adam: A Method for Stochastic Optimization”. In: *ICLR 2015*. 2015. DOI: 10.48550/arXiv.1412.6980. URL: <https://doi.org/10.48550/arXiv.1412.6980>.
- [54] Sergey Ioffe and Christian Szegedy. “Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift”. In: *ICML 2015*. 2015, pp. 448–456.
- [55] N. Srivastava et al. “Dropout: A Simple Way to Prevent Neural Networks from Overfitting”. In: *Journal of Machine Learning Research* 15 (2014), pp. 1929–1958.

- [56] Vinod Nair and Geoffrey E. Hinton. “Rectified Linear Units Improve Restricted Boltzmann Machines”. In: *ICML 2010*. 2010, pp. 807–814.
- [57] Dan Hendrycks and Kevin Gimpel. “Gaussian Error Linear Units (GELUs)”. In: *arXiv preprint arXiv:1606.08415* (2016). DOI: 10.48550/arXiv.1606.08415. URL: <https://doi.org/10.48550/arXiv.1606.08415>.
- [58] Bianca Zadrozny and Charles Elkan. “Transforming Classifier Scores into Accurate Multiclass Probability Estimates”. In: *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 2002, pp. 694–699. DOI: 10.1145/775047.775151. URL: <https://doi.org/10.1145/775047.775151>.
- [59] Meelis Kull, Telmo Silva Filho, and Peter Flach. “Beyond Temperature Scaling: Obtaining Well-Calibrated Multiclass Probabilities with Dirichlet Calibration”. In: *Advances in Neural Information Processing Systems (NeurIPS 2019)*. 2019. DOI: 10.48550/arXiv.1910.12675. URL: <https://doi.org/10.48550/arXiv.1910.12675>.
- [60] Mahdi Pakdaman Naeini, Gregory F. Cooper, and Milos Hauskrecht. “Obtaining Well-Calibrated Probabilities Using Bayesian Binning”. In: *Proceedings of the 29th AAAI Conference on Artificial Intelligence (AAAI 2015)*. 2015, pp. 2901–2907.
- [61] Tom Fawcett. “An Introduction to ROC Analysis”. In: *Pattern Recognition Letters* 27.8 (2006), pp. 861–874. DOI: 10.1016/j.patrec.2005.10.010. URL: <https://doi.org/10.1016/j.patrec.2005.10.010>.
- [62] Takaya Saito and Marc Rehmsmeier. “The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets”. In: *PLoS ONE* 10.3 (2015), e0118432. DOI: 10.1371/

- journal.pone.0118432. URL: <https://doi.org/10.1371/journal.pone.0118432>.
- [63] Glenn W. Brier. “Verification of Forecasts Expressed in Terms of Probability”. In: *Monthly Weather Review* 78.1 (1950), pp. 1–3. DOI: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2. URL: [https://doi.org/10.1175/1520-0493\(1950\)078%3C0001:VOFEIT%3E2.0.CO;2](https://doi.org/10.1175/1520-0493(1950)078%3C0001:VOFEIT%3E2.0.CO;2).
- [64] Lawrence D. Brown, T. Tony Cai, and Anirban DasGupta. “Interval Estimation for a Binomial Proportion”. In: *Statistical Science* 16.2 (2001), pp. 101–133. DOI: 10.1214/ss/1009213286. URL: <https://doi.org/10.1214/ss/1009213286>.
- [65] Bradley Efron and Robert Tibshirani. *An Introduction to the Bootstrap*. Boca Raton, FL: CRC Press, 1994.
- [66] Frank Wilcoxon. “Individual Comparisons by Ranking Methods”. In: *Biometrics Bulletin* 1.6 (1945), pp. 80–83.
- [67] X. Liu et al. “CONSORT-AI: Extension for Clinical Trial Reports for Interventions Involving Artificial Intelligence”. In: *BMJ* 370 (2020), p. m3164. DOI: 10.1136/bmj.m3164. URL: <https://doi.org/10.1136/bmj.m3164>.
- [68] S. C. Rivera et al. “SPIRIT-AI: Guidelines for Clinical Trial Protocols for Interventions Involving AI”. In: *BMJ* 370 (2020), p. m3210. DOI: 10.1136/bmj.m3210. URL: <https://doi.org/10.1136/bmj.m3210>.
- [69] P. M. Bossuyt et al. “STARD 2015: An Updated List of Essential Items for Reporting Diagnostic Accuracy Studies”. In: *BMJ* 351 (2015), h5527. DOI: 10.1136/bmj.h5527. URL: <https://doi.org/10.1136/bmj.h5527>.
- [70] V. Sounderajah et al. “STARD-AI: Standards for Reporting of Diagnostic Accuracy Studies that Use AI—Protocol”. In: *BMJ Open* 11.6 (2021),

- e047709. DOI: 10.1136/bmjopen-2020-047709. URL: <https://doi.org/10.1136/bmjopen-2020-047709>.
- [71] European Union. *Regulation (EU) 2016/679 (General Data Protection Regulation)*. Official Journal of the European Union, L119, pp. 1–88. 2016. URL: <https://eur-lex.europa.eu/eli/reg/2016/679/oj>.
- [72] A. Balapure et al. “Early Detection of Oral Cancer Biomarkers Using Microfluidic Devices: A Review”. In: *Analytical Methods* 16 (2024), pp. 233–248. DOI: 10.1039/D3AY02045A. URL: <https://doi.org/10.1039/D3AY02045A>.
- [73] Jianfeng Gao, Michel Galley, and Lihong Li. “Neural Approaches to Conversational AI”. In: *ACL 2018 Tutorial Abstracts*. 2018, pp. 2–7. DOI: 10.18653/v1/P18-5002. URL: <https://doi.org/10.18653/v1/P18-5002>.
- [74] Pranshu Goyal. *Synthetic Biomarker Training Table for DLQR (v1.0)*. Unpublished dataset; available from the author upon reasonable request. Galway, Ireland, 2025.
- [75] Pranshu Goyal. *Literature-Parameterised Hybrid Biomarker Matrix for DLQR (v1.0)*. Unpublished dataset; includes parameter tables (mean, SD, n) and generation scripts; available from the author upon reasonable request. Galway, Ireland, 2025.
- [76] Pranshu Goyal. *CBR Case Library (Numeric + Narrative) for DLQR (v1.0)*. Unpublished dataset; 10,000 cases (8,000 indexed/2,000 hold-out); available from the author upon reasonable request. Galway, Ireland, 2025.
- [77] Pranshu Goyal. *Clinical-Style Text Index for RAG (v1.0)*. Unpublished dataset; 30,636 passages with embeddings and index metadata; available from the author upon reasonable request. Galway, Ireland, 2025.

- [78] Pranshu Goyal. *Router Corpus and Decision/Calibration Logs for DLQR (v1.0)*. Unpublished dataset; available from the author upon reasonable request. Galway, Ireland, 2025.
- [79] P. Adusumilli et al. “Estimation of salivary and serum CYFRA 21-1 levels in patients with oral squamous cell carcinoma”. In: *Journal of Oral and Maxillofacial Pathology* 27.1 (2023), pp. 98–102. URL: https://journals.lww.com/jpat/fulltext/2023/27010/estimation_of_salivary_and_serum_cyfra_21_1.18.aspx.
- [80] Y.-H. Park et al. “Interleukin-6 as a prognostic biomarker in oral squamous cell carcinoma: A cohort analysis”. In: *International Journal of Molecular Sciences* 20.2 (2019), p. 332. DOI: 10.3390/ijms20020332.
- [81] I.-H. Chen et al. “Using SCC antigen and C-reactive protein levels as prognostic biomarkers in recurrent oral cavity squamous cell carcinoma”. In: *PLOS ONE* 9.7 (2014), e103265. DOI: 10.1371/journal.pone.0103265.
- [82] S.-F. Tai et al. “Roles of preoperative C-reactive protein are more relevant in buccal cancer than other subsites”. In: *World Journal of Surgical Oncology* 15.47 (2017). DOI: 10.1186/s12957-017-1116-5.
- [83] P. Kumari et al. “Expression of miR-31 in saliva-liquid biopsy in patients with oral squamous cell carcinoma”. In: *Saudi Dental Journal* 33.4 (2021), pp. 218–224. DOI: 10.1016/j.sdentj.2020.08.004.
- [84] C.-J. Liu et al. “miR-31-5p in saliva is a potential biomarker for oral squamous cell carcinoma”. In: *Head & Neck* 34.2 (2012), pp. 219–224. DOI: 10.1002/hed.21819.
- [85] K. B. Jadhav et al. “Expression of microRNA-21 in saliva and tumour tissue of patients with oral squamous cell carcinoma: A predictor of cervical lymph

node metastasis". In: *Oral Surgery, Oral Medicine, Oral Pathology and Oral Radiology* 133.1 (2022), pp. 60–69. DOI: 10.1016/j.oooo.2021.07.012.

Appendix A

Example Router I/O Transcripts

Listing A — Summarisation

```
{
  "trace_id": "ex-001",
  "route": "summarize",
  "result": {
    "bullets": [
      "Clinical records reflect oral cancer presentations (non-healing ulcers/masses).",
      "Ages ~32-81; mixed ethnicities.",
      "Biomarker elevations across saliva/blood/urine ... (e.g., IL-6, IL-8, CYFRA21-1, S",
      "Renal markers (creatinine/urea) often elevated.",
      "Exam notes ulcerative/exophytic lesions; biopsy/FNA used for work-up."
    ]
  }
}
```

Listing B — QA

```
{
  "trace_id": "ex-002",
```

```
"route": "qa",
"result": {
  "answer": "Cisplatin commonly causes nausea/vomiting, appetite loss, nephrotoxicity
(raised creatinine), ototoxicity (hearing loss or tinnitus), peripheral neuropathy,
mucositis, anaemia and hair loss; it may slightly increase secondary-cancer risk."
}
}
```

Listing C — Predict (tabular features)

```
{
  "trace_id": "ex-003",
  "route": "predict",
  "result": {
    "prediction": "OSCC",
    "confidence": 0.9997,
    "class_probabilities": {
      "OSCC": 0.9997,
      "Benign_Lesion": 0.0003,
      "Healthy_Control": 0.0
    }
  }
}
```

Appendix B

GitHub Repository

The complete implementation, evaluation harness, and research artifacts for this thesis are openly available in the public repository DLQR-Health. Same for archival reproducibility, the exact version used to produce all results.

How to obtain and verify.

```
git clone https://github.com/PranshuGoyal53/DLQR-Health.git
cd DLQR-Health
git checkout v1.0.0-thesis
```

Reproducibility notes. All experiments can be reproduced by checking out the archival tag above and following the repository’s setup instructions (e.g., creating a clean Python environment and installing dependencies as specified). Where randomness is involved, the seeds are set in code to enable repeatable runs.

Appendix C

Model Architectures

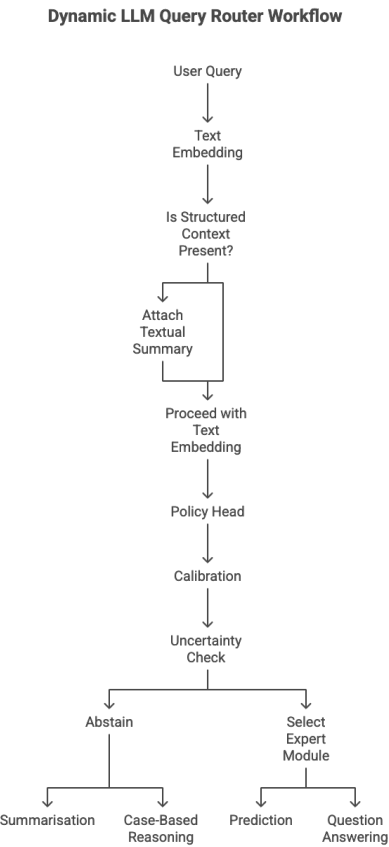


Figure C.1: DLQR Model Architectures

Appendix D

QA and Summarization Execution

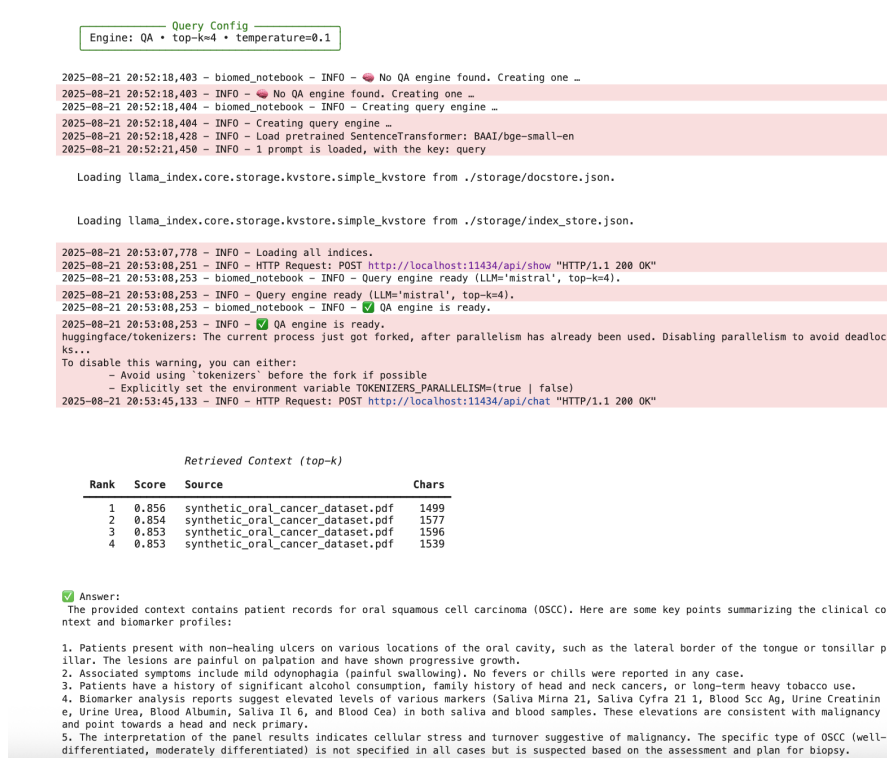


Figure D.1: QA Execution Individually

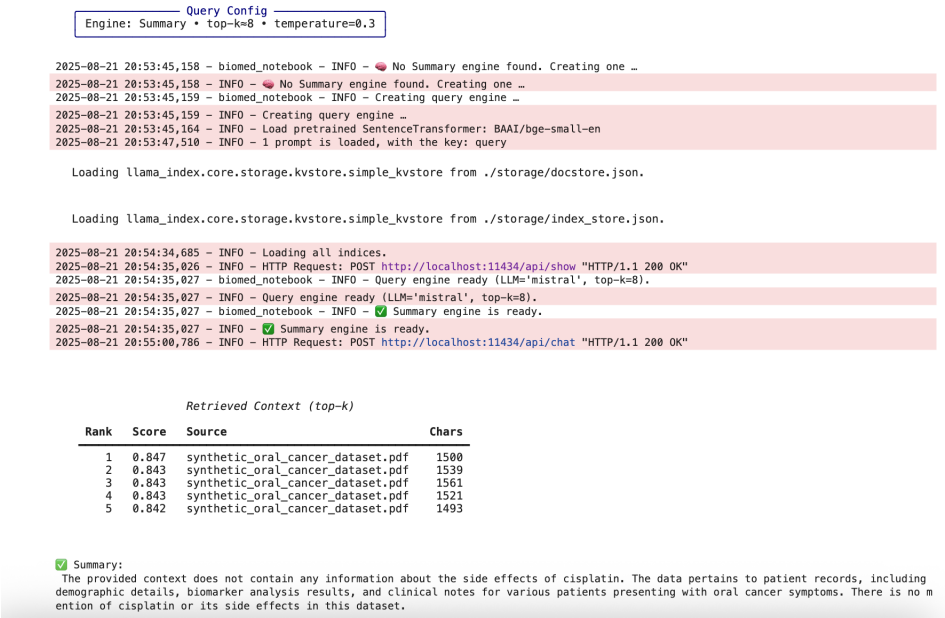


Figure D.2: Summarization Execution Individually